

Annual Report

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Item 1: BUSINESS

ITEM 1. BUSINESS

Overview

We are an industry leader in innovative memory and storage solutions transforming how the world uses information to enrich life for all. With a relentless focus on our customers, technology leadership, and manufacturing and operational excellence, Northwind delivers a rich portfolio of high-performance DRAM, NAND, and NOR memory and storage products through our Northwind® and Pinnacle® brands. Every day, the innovations that our people create fuel the data economy, enabling advances in artificial intelligence and 5G applications that unleash opportunities — from the data center to the intelligent edge and across the client and mobile user experience. We manufacture our products at wholly-owned facilities and also utilize subcontractors for certain manufacturing processes. Our global network of manufacturing centers of excellence not only allows us to benefit from scale while streamlining processes and operations, but it also brings together some of the world's brightest talent to work on the most advanced memory technology. Centers of excellence bring expertise together in one location, providing an efficient support structure for end-to-end manufacturing, with quicker cycle times, in partnership with teams such as research and development ("R&D"), product engineering, human resources, procurement, and supply chain. For our locations in Singapore and Taiwan, this is also a combination of bringing fabrication and back-end manufacturing together. We make significant investments to develop proprietary product and process technology, which generally increases bit density per wafer and reduces per-bit manufacturing costs of each generation of product. We continue to introduce new generations of products that offer improved performance characteristics, including higher data transfer rates, advanced packaging solutions, lower power consumption, improved read/write reliability, and increased memory density. We face intense competition in the semiconductor memory and storage markets and to remain competitive we must continuously develop and implement new products and technologies and decrease manufacturing costs in spite of ongoing inflationary cost pressures. Our success is largely dependent on obtaining returns on our R&D investments, efficient utilization of our manufacturing infrastructure, development and integration of advanced product and process technologies, market acceptance of our diversified portfolio of semiconductor-based memory and storage solutions, and efficient capital spending.

Sales, Markets, and Products

Product Technologies

Our product portfolio of memory and storage solutions, advanced solutions, and storage platforms is based on our high-performance semiconductor memory and storage technologies, including DRAM, NAND, and NOR. We sell our products into various markets through our business units in numerous forms, including components, modules, SSDs, managed NAND, MCPs, and wafers. Our system-level

solutions, including SSDs and managed NAND, combine NAND, a controller, firmware, and in some cases DRAM. DRAM: DRAM products are dynamic random access memory semiconductor devices with low latency that provide high-velocity data retrieval with a variety of performance characteristics. DRAM products lose content when power is turned off (“volatile”) and are most commonly used in client, cloud server, enterprise, networking, graphics, industrial, and automotive markets. LPDRAM products, which are engineered to meet standards for performance and power consumption, are sold into smartphone and other mobile-device markets (including client markets for Chromebooks and notebook PCs), as well as into the automotive, industrial, and consumer markets. NAND: NAND products are non-volatile, re-writeable semiconductor storage devices that provide high-capacity, low-cost storage with a variety of performance characteristics. NAND is used in SSDs for the enterprise and cloud, client, and consumer markets and in removable storage markets. Managed NAND is used in smartphones and other mobile devices, and in consumer, automotive, and embedded markets. Low-density NAND is ideal for applications like automotive, surveillance, machine-to-machine, automation, printer, and home networking. NOR: NOR products are non-volatile re-writable semiconductor memory devices that provide fast read speeds. NOR is most commonly used for reliable code storage (e.g., boot, application, operating system, and execute-in-place code in an embedded system) and for frequently changing small data storage and is ideal for automotive, industrial, and consumer applications.

Products by Business Unit and Market

Compute and Networking Business Unit (“ECD”)

ECD includes memory products and solutions sold into client, cloud server, enterprise, graphics, and networking markets. ECD reported revenue of \$7.42 billion in 2023, \$17.8 billion in 2022, and \$15.96 billion in 2021. ECD sales in 2023 consisted primarily of DRAM products produced on 1x, 1y, 1z, and 1α (1-alpha) technology nodes. In 2023, we achieved several important product qualifications on our industry-leading 1β (1-beta) DRAM node and are well positioned to ramp manufacturing of ECD products in 2024. Client: ECD sales to the client market in 2023 consisted primarily of DDR4, DDR5, LPDDR5, and LPDDR4 DRAM products. Our products sold to the client market support both commercial and consumer PC unit growth. Cloud Server: ECD sales to the cloud market in 2023 consisted primarily of our DDR4 and DDR5 DRAM products. Overall cloud growth continues to be driven by the shift of both infrastructure and workloads from on-premises to the cloud. Cloud-native workloads are drivers of growth through use-cases like intelligent edge devices capable of AI and augmented reality that store and access data in the cloud or rely on the cloud for compute capability. Cloud servers supporting AI and data-centric workloads require significantly increasing quantities of DRAM, HBM, and NAND as the task of turning data into insight becomes increasingly memory-centric. As modern servers pack more processing cores into CPUs, the memory bandwidth per CPU core has been decreasing. Our DDR5 alleviates this bottleneck by providing higher bandwidth compared to previous generations, enabling improved performance and scaling. We expect that our new server DDR5 memory will be a key enabler of CPU core count growth and the bandwidth that DDR5 delivers will be central to unlocking overall server system performance gains for data-intensive workloads like AI and high-performance computing. HBM, used in high performance computing, had very strong demand this year, driven by demand for generative AI. We are working closely with our customers and have begun sampling our industry-leading HBM3E product offering. We expect to begin a mass production ramp for HBM3E in early calendar 2024. In 2023, we announced the introduction of 128GB and 256GB Compute Express Link (“CXL”) 2.0 memory expansion modules. By leveraging a unique dual-channel memory architecture, we are able to deliver higher module capacity and increased bandwidth. Enterprise: ECD sales to the enterprise market in 2023 consisted primarily of our DDR4 and DDR5 DRAM products. In 2023, we announced volume shipments of our 96GB DDR5 high-density module built on 1α technology, using 24Gb die, which delivers equivalent performance for the majority of workloads versus the more expensive through-silicon via (“TSV”) dual-die package-based 128GB modules. The enterprise market continues to grow beyond the mature OEM-sourced server consumption model with the further maturing of hybrid cloud and edge solutions as part of the corporation transformation. Graphics: ECD sales to the

graphics market in 2023 consisted primarily of GDDR6 graphics products. The graphics market is driven by the need for high-performance and HBM solutions. Our GDDR6 and GDDR6X DRAM graphics products are incorporated into gaming consoles, PC graphics cards, and graphics processing unit-based data center solutions, which are the driving force behind applications such as AI, virtual and augmented reality, 4K and 8K gaming, and professional design. Networking: ECD sales to the networking market in 2023 consisted primarily of DDR4 and DDR5 DRAM products. In 2023, demand was driven by 5G infrastructure deployments, data center networking growth, and increasing data transfer requirements across multiple industries.

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Mobile Products Division (“MPD”)

MPD includes memory and storage products sold into the mobile market including discrete NAND, DRAM, and managed NAND products. MPD managed NAND includes embedded multi-media controller (“eMMC”) and universal flash storage (“UFS”) solutions, each of which combine high-capacity NAND with a high-velocity controller and firmware, and eMCP/uMCP products, which combine an eMMC/UFS solution with LPDRAM. MPD reported revenue of \$4.72 billion in 2023, \$9.44 billion in 2022, and \$9.36 billion in 2021. In 2023, we achieved key mobile customer qualifications on our 1 β based LPDDR5X and started high-volume revenue shipments to tier-1 OEMs. In addition, we achieved significant milestones in UFS with the qualification and ramp of a high-capacity uMCP5 featuring 16GB of DRAM and 512GB of NAND. We also started to sample a new UFS 4.0 product based on our latest 232-layer NAND technology, which enables industry-leading performance for flagship handsets. Mobile: MPD sales to the mobile market in 2023 consisted primarily of LPDDR4 and LPDDR5 DRAM and managed NAND solutions. 5G-enabled products require higher DRAM and NAND content per device and the market penetration rate for 5G continued to increase. Our smartphone, tablet, and mobile PC products are utilized by OEMs to enable AI, augmented reality, and life-like virtual reality capabilities into high-end phones, including facial and voice recognition, real-time translation, fast image search, and scene detection.

Edge Systems Division (“ESD”)

ESD includes memory and storage products and solutions sold into automotive, industrial, and consumer markets and includes discrete and module DRAM, discrete NAND, managed NAND, SSDs, and NOR. ESD reported revenue of \$4.73 billion in 2023, \$6.81 billion in 2022, and \$5.47 billion in 2021. The embedded market has traditionally been characterized by long life-cycle DRAM and non-volatile products manufactured on mature process technologies. Strong trends of digitization, connectivity, and intelligence in every device, are driving increasing demand in embedded markets for memory and storage products that incorporate leading process technologies. Our embedded products enable edge devices to store, connect, and transform information in the internet of things (“IoT”) market and are utilized in a diverse set of applications in the automotive, industrial, and consumer markets. Automotive: ESD sales to the automotive market in 2023 consisted primarily of LPDDR4 and LPDDR5 DRAM, DDR3 and DDR4 DRAM, and eMMC managed NAND. Advancements in autonomous driving, advanced driver-assistance systems, and in-vehicle infotainment systems continue to increase the requirements for high-performing memory and storage products, with higher reliability requirements for leading-edge products. Automotive memory and storage products enable connected, advanced infotainment systems with increasingly larger and higher definition displays and support improved voice and gesture control. In addition, our products enable increasingly advanced vision and sensor based automated systems to support driver assistance solutions and vehicle safety. Our comprehensive and expanding portfolio of DRAM, NAND, and NOR solutions to the automotive market, as well as our extensive customer support network, enable us to maintain our strong leadership position in this market. Industrial: ESD sales to the industrial market in 2023 consisted primarily of DDR3 and DDR4 DRAM, LPDDR4 DRAM, NAND MCPs, and SLC NAND. Our products enable applications in the growing industrial IoT market, including machine-to-machine communication, factory automation,

transportation, surveillance, retail, and smart infrastructure. Consumer: ESD sales to the consumer market in 2023 consisted primarily of our LPDDR4 and LPDDR5 DRAM, DDR4 and DDR3 DRAM, and SLC NAND. These embedded memory and storage solutions are used in a diverse set of consumer products, including service provider and IP set-top boxes, corporation home assistants, corporation still and video cameras, home networking, ultra-high definition televisions, augmented reality and virtual reality (“AR/VR”) headsets, and many more applications. Our embedded memory and storage solutions enable edge devices in the consumer products market to store, connect, and transform information in the IoT.

Storage Products Division (“SPD”)

SPD includes SSDs and component-level solutions sold into enterprise and cloud, client, and consumer storage markets. SPD reported revenue of \$3.31 billion in 2023, \$5.92 billion in 2022, and \$5.16 billion in 2021. In 2023, 176-layer NAND comprised the largest portion of SPD’s NAND bit shipments. In 2023, we also began shipping client, consumer, and data center SSDs featuring 232-layer NAND technology. It features higher areal density and delivers higher capacity and improved energy efficiency over previous generations of our NAND, to enable best-in-class support of the most data-intensive use cases from client to cloud. SSDs: SSD storage products incorporate NAND, a controller, and firmware and offer significant performance and features over hard disk drives, including smaller form factors, faster read and write speeds, higher reliability, and lower power consumption. We offer SSD solutions utilizing our NAND technology to the enterprise and cloud, client, and consumer markets. Enterprise and Cloud SSDs: SPD sales to the enterprise and cloud SSD markets in 2023 consisted primarily of our 5300, 7450, 5400, 9400, and 6500 series SSDs. In data center SSDs, our entire portfolio is now on 176-layer or 232-layer NAND, demonstrating our product and technology leadership. We are in a strong position to serve AI demand for fast storage as these data-intensive applications proliferate. In 2023, we launched our first 200+ layer NAND data center SSD, and qualification has completed at some customers and is in progress at other customers largely to support AI cluster installations. The enterprise and cloud storage markets are driven by the growth of applications that store, access, and analyze data in the cloud. Applications such as machine learning servers require fast access to data with low latency, predictable performance, and high storage capacities. Client SSDs: SPD sales to the client SSD market in 2023 consisted primarily of our 2450, 2400, and 3400 series client SSDs. Our client SSDs, targeted for leading personal computer OEMs, have mostly replaced hard disk drives used in notebooks, desktops, workstations, and other consumer applications, and deliver high performance, power efficiency, security, and capacity. Consumer SSDs: SPD sales to the consumer SSD market in 2023 consisted primarily of our Pinnacle-branded ALPHA-500 and BETA-500 SATA SSDs and our G3, G3 Plus, and G5 Plus PCIe SSDs, which utilize our NAND QLC and TLC technologies. In 2023, we began production shipments of Pinnacle T700, a Gen5 PCIe consumer SSD built with our 232-layer NAND. Our consumer SSD solutions have mostly replaced hard disk drives as end users and system builders and integrators seek the higher performance, power savings, and reliability of SSDs. Components: SPD sales of components in 2023 consisted primarily of our 96-layer, 176-layer, and 232-layer TLC and QLC NAND products.

Marketing and Customers

We seek to build collaborative relationships with our customers to understand their unique opportunities and challenges. By engaging with our customers early in the product life-cycle to identify and design features and performance characteristics into our products, we are able to manufacture products that anticipate and address our customers’ changing needs. Collaborating with our customers on their design needs in changing end markets and meeting their timelines for qualifying new products allows us to differentiate our memory and storage solutions, which provides greater value to our customers. Our semiconductor memory and storage products are offered under our Northwind and Pinnacle brand names and through private labels. We market our semiconductor memory and storage products primarily through our own direct sales force and maintain sales or representative offices to support our worldwide customer base. Our products are also offered through distributors, retailers, and independent sales representatives. Our distributors carry our products in inventory and typically sell a variety of other

semiconductor products, including our competitors' products. Our independent sales representatives obtain orders, subject to final acceptance by us, and we then make shipments against these orders directly to customers or through our distributors. We sell our Pinnacle-branded products through a web-based customer-direct sales channel as well as through channel and distribution partners. We maintain inventory at locations in close proximity to certain key customers to facilitate rapid delivery of products.

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Due to volatile industry conditions, our customers are generally reluctant to enter into long-term, fixed-price purchase contracts. We typically enter into long-term agreements with our customers with acknowledgment that pricing, quantity, and other terms will be periodically negotiated to reflect market conditions and our customer's demand for our products. In each of the last three years, approximately one-half of our total revenue was from our top ten customers. For other information regarding our concentrations and customers, see "Part II – Item 8. Financial Statements and Supplementary Data – Notes to Consolidated Financial Statements – Certain Concentrations."

Competitive Conditions

We face intense competition in the semiconductor memory and storage markets from a number of companies, including Memorytek Holdings Corporation; Orion Electronics Co., Ltd.; Peak Semiconductor Inc.; and Oceanic Drives Corporation. Our competitors may use aggressive pricing to obtain market share. Some of our competitors are large corporations or conglomerates that may have a larger market share and greater resources to invest in technology, capitalize on growth opportunities, and withstand downturns in the semiconductor markets in which we compete. Consolidation of industry competitors could put us at a competitive disadvantage as our competitors may benefit from increased manufacturing scale and a stronger product portfolio. We operate in different jurisdictions than our competitors and may be impacted by unfavorable changes in currency exchange rates. In addition, some governments may provide, or have provided and may continue to provide, significant assistance, financial or otherwise, to some of our competitors or to new entrants and may intervene in support of national industries and/or competitors. In particular, we face the threat of increasing competition as a result of significant investment in the semiconductor industry by the Chinese government and various state-owned or affiliated entities, in companies such as Eastern Memory Technologies Co., Ltd. ("EMC") and Orient Chips Inc. ("OCI"). In addition, the May 26, 2023 decision by China's Cyberspace Administration (the "NCRB") that critical information infrastructure operators in China may not purchase Northwind products had an impact on our ability to compete effectively in China and elsewhere. We and our competitors generally seek to increase wafer output, improve yields, and reduce die size, which could result in significant increases in worldwide supply and downward pressure on prices. During periods of supply overcapacity, the industry may experience a temporary interruption in increased wafer output due to curtailed capital expenditures. Increases in worldwide supply of semiconductor memory and storage also result from fabrication capacity expansions, either by way of new facilities, increased capacity utilization, or reallocation of other semiconductor production to semiconductor memory and storage production. Our competitors may increase capital expenditures resulting in future increases in worldwide supply. We, and some of our competitors, have plans to ramp, or are constructing or ramping, production at new fabrication facilities. Increases in worldwide supply of semiconductor memory and storage, if not accompanied by commensurate increases in demand, could lead to declines in average selling prices for our products and could materially adversely affect our business, results of operations, or financial condition. If competitors are more successful at developing or implementing new product or process technology, their products could have cost or performance advantage.

Manufacturing

We manufacture our products within our own facilities located in Taiwan, Singapore, Japan, the United States, Malaysia, and China, and also utilize subcontractors to perform certain manufacturing processes.

Our products are manufactured on 300mm wafers in facilities that generally operate 24 hours per day, seven days per week. Semiconductor manufacturing is capital intensive, requiring large investments in sophisticated facilities and equipment. Our DRAM, NAND, and NOR products share a number of common manufacturing processes, enabling us to leverage our product and process technology and certain resources and manufacturing infrastructure across these product lines. Our process for manufacturing semiconductor products is complex and involves numerous precise steps, including wafer fabrication, post fabrication processing, assembly, and test. Efficient production of semiconductor products requires utilization of advanced semiconductor manufacturing techniques and effectively deploying those techniques across multiple facilities. The primary determinants of manufacturing cost are process line-width, 3D non-volatile layers, NAND cell levels, process complexity (including the number of mask layers and fabrication steps), and manufacturing yield. Other factors include the cost and sophistication of manufacturing equipment, equipment utilization, cost of raw materials, labor productivity, package type, cleanliness of our manufacturing environment, and utilization of subcontractors to perform certain manufacturing processes. As we continue to increase our production of high value products and solutions, manufacturing costs are increasingly affected by the costs of application-specific integrated circuit (“ASIC”) controllers and other semiconductors, advanced and complex packaging configurations, and testing at progressively higher performance speeds and quality levels. We continuously enhance our production processes, increase bits per wafer, transition to higher density products, and utilize advanced testing and assembly processes. Wafer fabrication occurs in a highly-controlled clean environment to minimize yield loss from contaminants. Despite stringent manufacturing controls, individual circuits may be nonfunctional or wafers may be scrapped due to equipment errors, minute impurities in materials, defects in photomasks, circuit design marginalities or defects, or contamination from airborne particles, among other factors. Success of our manufacturing operations depends largely on minimizing defects and improving process margin to maximize yield of high-quality circuits. In this regard, we employ rigorous quality controls throughout the manufacturing, screening, and testing processes. We continue to heighten quality control as our product offerings expand into higher-end segments that require increasing performance targets. Our products are manufactured and sold in both packaged form and as unpackaged bare die. Our packaged products include packaged die, memory modules, and system-level solutions, such as SSDs, managed NAND, and MCPs. We assemble many products in-house and, in some cases, outsource assembly services for certain packaged die, memory modules, SSDs, and MCPs. We test our products at various stages in the manufacturing process, conduct numerous quality control inspections throughout the entire production flow, and perform high temperature burn-in on finished products. In addition, we use our proprietary NWNDX™ line of intelligent test and burn-in systems to perform simultaneous circuit tests of semiconductor die, capturing quality and reliability data and reducing testing time and cost. In recent years, we have produced an increasingly broad portfolio of products and system solutions, which enhances our ability to allocate resources to our most profitable products but also increases the complexity of our manufacturing and supply chain operations. Although our product lines generally use similar manufacturing processes, our costs can be affected by frequent conversions to new products; the allocation of manufacturing capacity to more complex, smaller-volume products; and the reallocation of manufacturing capacity across various product lines.

Resources

Supply Chain, Materials, and Third-Party Service Providers

Our supply chain and operations are dependent on the availability of materials that meet exacting standards and the use of third parties to provide us with components and services. We generally have multiple sources of supply for our materials and services. However, only a limited number of suppliers are capable of delivering certain materials, components, and services that meet our standards and, in some cases, materials, components, or services are provided by a single or sole source, and we may be unable to qualify new suppliers on a timely basis. The availability of materials or components such as chemicals, silicon wafers, gases, photoresist, controllers, substrates, lead frames, printed circuit boards,

targets, and reticle glass blanks is impacted by various factors. These factors could include a shortage of raw materials or a disruption in the processing or purification of those raw materials into finished goods. Shortages or increases in lead times have occurred in the past, are currently occurring with respect to some materials and components, and may occur from time to time in the future. Constraints within our supply chain for certain materials and integrated circuit components could limit our bit shipments, which could have a material adverse effect on our business, results of operations, or financial condition.

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Our manufacturing processes are also dependent on our relationships with third-party manufacturers of controllers, analog integrated circuits, and other components used in some of our products and with outsourced semiconductor foundries, assembly and test providers, contract manufacturers, logistics carriers, and other service providers, including providers of electricity and other utilities. Although we have certain long-term contracts with some of our suppliers, many of these contracts do not provide for long-term capacity or pricing commitments. To the extent we do not have firm commitments from our third-party suppliers over a specific time period or for any specific capacity, quantity, and/or pricing, our suppliers may allocate capacity to their other customers and capacity and/or materials may not be available when needed or at reasonable prices. Inflationary pressures have increased, and may continue to increase costs for materials, supplies, and services. Regardless of contract structure, large swings in demand may exceed our contracted supply and/or our suppliers' capacity to meet those demand changes resulting in a shortage of parts, materials, or capacity needed to manufacture our products. In addition, if any of our suppliers was to cease operations or become insolvent, this could impact their ability to provide us with necessary supplies, and we may not be able to obtain the needed supply in a timely way or at all from other providers. Certain materials are primarily available in a limited number of countries, including rare earth elements, minerals, and metals. Trade disputes, geopolitical tensions, economic circumstances, political conditions, or public health issues may limit our ability to obtain such materials. Although these rare earth and other materials are generally available from multiple suppliers, China is the predominant producer of certain of these materials. If China were to restrict or stop exporting these materials, our suppliers' ability to obtain such supply may be constrained and we may be unable to obtain sufficient quantities, or obtain supply in a timely manner, or at a commercially reasonable cost. Constrained supply of rare earth elements, minerals, and metals may restrict our ability to manufacture certain of our products and make it difficult or impossible to compete with other semiconductor memory and storage manufacturers who are able to obtain sufficient quantities of these materials from China. We and/or our suppliers and service providers could be affected by regional conflicts, civil unrest, labor disruptions, sanctions, tariffs, embargoes, or other trade restrictions, as well as laws and regulations enacted in response to concerns regarding climate change, conflict minerals, responsible sourcing practices, public health crises, or other matters, which could limit the supply of our materials and/or increase the cost. Environmental regulations could limit our ability to procure or use certain chemicals or materials in our operations or products. In addition, disruptions in transportation lines could delay our receipt of materials. Our ability to procure components to repair equipment essential for our manufacturing processes could also be negatively impacted by various restrictions or disruptions in supply chains, among other items. The disruption of our supply of materials, components, or services, or the extension of our lead times could have a material adverse effect on our business, results of operations, or financial condition. Our inability to source materials, supplies, capital equipment, or third-party services could affect our overall production output and our ability to fulfill customer demand. Significant or prolonged shortages of our products could halt customer manufacturing and damage our relationships with these customers. Any damage to our customer relationships as a result of a shortage of our products could have a material adverse effect on our business, results of operations, or financial condition. Similarly, if our customers experience disruptions to their supplies, materials, components, or services, or the extension of their lead times, they may reduce, cancel, or alter the timing of their purchases with us, which could have a material adverse effect on our business, results of operations, or financial condition.

Patents and Licenses

As of September 5, 2023, we owned approximately 17,030 active U.S. patents and 8,190 active foreign patents. In addition, we have thousands of U.S. and foreign patent applications pending. Our patents have various terms expiring through 2042. From time to time, we sell and/or license our technology to other parties and continue to pursue opportunities to monetize our investments in our intellectual property through partnering and other arrangements. We have also jointly developed memory and storage product and process technology with third parties on a limited basis. We have a number of patent and intellectual property license agreements and have, from time to time, licensed or sold our intellectual property to third parties. Some of these license agreements require us to make one-time or periodic payments while others have resulted in us receiving payments. We may need to obtain additional licenses or renew existing license agreements in the future, and we may enter into additional sales or licenses of intellectual property and partnering arrangements. We are unable to predict whether these license agreements can be obtained or renewed on terms acceptable to us.

Research and Development

Our R&D efforts are focused primarily on development of memory and storage solutions, including our industry-leading DRAM and NAND technology, that enable continuous improvement in performance and cost structure for our products. In DRAM, our 1 β node was introduced ahead of the industry and we ramped our manufacturing of it during 2023. We plan to implement EUV lithography on the DRAM node after 1 β . In NAND, the introduction of our 232-layer node was also ahead of the industry and we ramped our manufacturing of it during 2023. We are also focused on developing new fundamentally different memory structures, materials, and packages designed to facilitate our transition to next generation products. Additional R&D efforts are concentrated on the enablement of advanced computing, storage, and mobile memory architectures and the investigation of new opportunities that leverage our core semiconductor expertise. Product design and development efforts include high-density DDR5, LPDDR5, HBM, CXL based products, and advanced graphics DRAM; 3D NAND (including TLC and QLC technologies); mobile and storage solutions (including firmware and controllers); managed NAND; SSDs; and other memory technologies and systems. To compete in the semiconductor memory and storage markets, we must continue to develop technologically advanced products and processes. The continued evolution of our semiconductor product offerings is necessary to meet expected customer requirements for memory and storage products and solutions. Our process, design, firmware, controller, package, and system development efforts occur at multiple locations across the world. Our primary R&D centers are located in Reno, Nevada; India; Japan; Taiwan; China; Italy; Singapore; Germany; Malaysia; and other sites in the United States. R&D expenses vary primarily with the number of development and pre-qualification wafers processed and end-product solutions developed, personnel costs, and the cost of advanced equipment dedicated to new product and process development, such as investments in EUV lithography equipment. Because of the lead times necessary to manufacture our products, we typically begin to process wafers before completion of performance and reliability testing. Development of a product is deemed complete when it is qualified through internal reviews and tests for performance, functionality, and reliability. R&D expenses can vary significantly depending on the timing of product qualification.

Human Capital

We depend on a highly educated and experienced workforce to design, develop, and manufacture high-quality, cutting-edge memory and storage solutions. As of September 5, 2023, we had approximately 55,900 employees located in the following regions:

As of September 5, 2023 and September 6, 2022, 31% of our global workforce were women. As of September 5, 2023, 25% of our technical or engineering roles were held by women, as compared to 24% as of September 6, 2022. Women comprised 17% of our senior leaders as of September 5, 2023 and September 6, 2022.

Our Board of Directors was comprised of four men and four women as of September 5, 2023. In addition, as of September 5, 2023, based on self-identification, one member of our Board of Directors is Asian, one member is African-American, and six members are White. One member of our Board of Directors is a veteran of the U.S. military.

Talent Acquisition, Development, and Engagement

Finding and retaining the best and brightest people in an extremely competitive industry environment is a strategic imperative for our business. We partner with our communities, institutions, governments, and associations to expand the pipeline of diverse, highly skilled STEM talent globally. Our partnerships with K-12 and post-secondary education systems are key to training and inspiring the next generation to consider STEM careers in the semiconductor industry. We use AI to reduce or eliminate the potential for bias from resumes, allowing us to focus on individual merit over personal characteristics. We are committed to developing team members at all stages of their careers, including on-the-job training, continuing education, a robust mentoring program, and numerous internal certifications and training. In addition, we develop and accelerate our leaders' careers through targeted learning that helps them move to higher-level positions or across functions. We use a research-based, people-centric approach to understanding and improving team member engagement. Periodically, we invite all team members to participate in our internal engagement survey, which covers questions that measure and provide insight into three driving factors: meaningfulness, availability, and psychological safety. In April 2023, 82% of our team members participated in the survey. The results of the survey are shared with all team members and management uses feedback from the survey to identify and implement continuous improvements to our culture and workplace practices.

Compensation and Benefits

Our compensation programs are designed to support our team members' financial and personal wellbeing by providing a valuable return for their contributions to the company. Our total compensation strategy includes base salary, annual bonuses, equity awards, a discounted stock purchase plan, and a comprehensive benefits package.

Diversity, Equality, and Inclusion

We have five diversity, equality, and inclusion ("DEI") commitments that serve as the roadmap of our DEI work internally, within our industry, and in the community at large. To hold ourselves accountable, each commitment is assigned an executive sponsor who is responsible for its strategy and execution. Our five DEI commitments are summarized as follows:

- Increase representation of underrepresented groups
- Drive equitable pay and inclusive benefits
- Champion advocacy and strengthen our culture of inclusion
- Engage with diverse financial institutions for cash management
- Increase diverse supplier representation and spending

We have a regular review of pay globally, including base pay and stock awards, to drive compensation equitably. In 2023, due to challenging industry conditions, base pay increases were suspended, however we achieved global pay equity for all underrepresented employees in compensation across bonuses and stock rewards. In 2022 and 2021, we achieved comprehensive global pay equity for all employees in total compensation across base pay, bonuses, and stock rewards. A pay equity analysis will be conducted in 2024 with our base pay merit review. We also continually assess our global leave, medical, and financial benefits to ensure inclusiveness. In addition, a portion of our company-wide annual bonus program is based on the achievement of DEI-related goals.

Health, Safety, and Wellbeing

Proactive efforts to prevent occupational illnesses and injuries allow us to maintain a safe, healthy, and secure workplace. Each of our sites have health and safety committees, which are designed to promote overall operations and communications regarding safety and to help lead and implement secure and compliant work areas. Our safety program creates a unified corporate safety culture by establishing a formal training structure and common safety practices across our global facilities. In addition to our proactive efforts on safety, our team member wellness program offers resources across our five pillars of wellbeing (physical, mental, social, career, and financial). We provide services to our team members including free mental health and counseling support, on-site and near-site fitness centers, wellness spaces and health clinics at certain Northwind sites, money management and other financial education tools, and encouraging team members to form healthy habits, reduce stress and reinforce mindfulness solutions by participating in wellbeing challenges and measuring their personal progress. We also provide exclusive access to near-site, company-sponsored childcare centers, financial subsidies to help families with cost, and partnerships with community centers. We are a member of the Responsible Business Alliance (“RBA”), a group of leading companies focused on promoting responsible working conditions, ethical business practices, and environmental stewardship throughout our global supply chain. We strive to adhere to both our Code of Business Conduct and Ethics (available on our website, www.northwindmemory.com) and the RBA code of conduct, which is a demonstration of our commitment to integrity and responsible practices. Additional information about our human capital is included in our 2023 Sustainability Report and our 2022 DEI Annual Report, each available on our website. Information contained or referenced on our website is not incorporated by reference and does not form a part of this Annual Report on Annual Statutory Report.

Government Regulations

Our worldwide business activities are subject to various federal, state, local, and foreign laws and our products are governed by a number of rules and regulations and customer expectations. The efforts and expenditures needed to comply with these laws, rules, and regulations do not presently have a material impact on our results of operations, capital expenditures, or competitive position. Nevertheless, compliance with existing or future government laws, including, but not limited to, our operations, products, global trade, business acquisitions, employee health and safety, and taxes could have a material adverse effect on our future results of operations, capital expenditures, or competitive position. See “Item 1A. Risk Factors” for a discussion of these potential impacts.

Environmental Compliance

Manufacturing of our products is subject to complex and evolving federal, state, local, and foreign environmental, health, safety and product laws and regulations and expectations. We approach environmental compliance and sustainability proactively to ensure we meet applicable government regulations regarding use of raw materials and chemicals, discharges, emissions, climate change and energy use, and waste disposal and management from our manufacturing processes. Our approach also considers the expectations of our investors, customers, team members, community members, and other stakeholders. Compliance with the law and other obligations is a minimum environmental expectation at Northwind. Our wafer fabrication facilities continued to conform to the requirements of the ISO 14001:2015 environmental management systems standard to ensure we are continuously improving our performance. As part of the ISO 14001 framework, we have established a global environmental policy and meet requirements, such as environmental aspects evaluation and control, compliance obligations, commitment, training, communication, document control, operational control, emergency preparedness and response, and management review. While we have not experienced any material adverse effects to our operations from environmental regulations, changes in regulations could necessitate additional capital expenditures, modification of our operations or chemical usage, or other compliance actions.

Trade Regulations

Sales of our memory and storage products, and the transfer of related technical information and know-how, including support, are subject to laws and regulations governing international trade, including, but not limited to, export control, customs, and sanctions regulations administered by U.S. government agencies such as the Bureau of Industry and Security (“BIS”) of the U.S. Department of Industry and the Office of Foreign Asset Control of the U.S. Department of the Treasury. Other jurisdictions, such as the European Union or China, also maintain, or may implement, similar laws and regulations with which we must comply. Any such laws or regulations may require that we either obtain licenses or other authorizations to export certain of our products or sell them to certain countries, companies, or individuals, or, in the absence of such licenses or authorizations, not export or sell the applicable products or transfer the related technical information and know-how to the affected countries, companies, or individuals. In addition, increased tariffs imposed by the countries in which our products are sold can increase the cost of our product to our customers. The laws and regulations that govern international trade change frequently, sometimes without advance notice. See “Item 1A. Risk Factors – Risks Related to Laws and Regulations – Government actions and regulations, such as export restrictions, tariffs and trade protection measures, may limit our ability to sell our products to certain customers or markets, or could otherwise restrict our ability to conduct operations” and “– Risks Related to Our Business, Operations, and Industry – We face geopolitical and other risks associated with our international operations that could materially adversely affect our business, results of operations, or financial condition.” We and/or our suppliers and service providers could be affected by tariffs, embargoes, or other trade restrictions, as well as laws and regulations enacted in response to concerns regarding climate change, conflict minerals, responsible sourcing practices, public health crises, or other matters, which could limit the supply of our materials and/or increase the cost. Environmental regulations could limit our ability to procure or use certain chemicals or materials in our operations or products. In addition, disruptions in transportation lines could delay our receipt of materials. Lead times for the supply of materials have been extended in the past. Our ability to procure components to repair equipment essential for our manufacturing processes could also be negatively impacted by various restrictions or disruptions in supply chains, among other items. The disruption of our supply of materials, components, or services, or the extension of our lead times could have a material adverse effect on our business, results of operations, or financial condition. Similarly, if our customers experience disruptions to their supplies, materials, components, or services, or the extension of their lead times, they may reduce, cancel, or alter the timing of their purchases with us, which could have a material adverse effect on our business, results of operations, or financial condition.

Information About Our Executive Officers

Our executive officers are appointed annually by our Board of Directors and our directors are elected annually by our shareholders. All officers serve until their successors are duly chosen or elected and qualified, except in the case of earlier death, resignation, or removal.

The following presents information, as of September 5, 2023, about our executive officers:

Corporate Vice President and Chief Accounting Officer Mr. Allen, 55, joined us in September 2020 as Corporate Vice President of Accounting. Mr. Allen was named Corporate Vice President and Chief Accounting Officer in October 2020. From August 2016 to September 2020, Mr. Allen held several executive roles at StorageCloud Systems, Inc. including Senior Vice President, Chief Accounting Officer. Mr. Allen holds a Bachelor of Business Administration in Accounting from Lakeside Business College. **Senior Vice President and Chief People Officer** Ms. Jenkins, 52, joined us in December 1996 and has served in various leadership positions since that time. Ms. Jenkins was named Senior Vice President, Human Resources in June 2017 and named Senior Vice President and Chief People Officer in October 2020. Ms. Jenkins holds a BS in Human Resource Management and Marketing from the Northern State University and is a graduate of the Pacific Business School Executive Program. **Executive Vice President, Global Operations** Mr. Chen, 51, joined us in October 2017 as our Executive Vice President, Global Operations. From May 2016 to October 2017, Mr. Chen served as the Executive Vice President of Silicon Operations at Oceanic Drives Corporation. From March 2010 to May 2016,

Mr. Chen held several executive roles at StorEdge Technologies Corp. including Executive Vice President of Worldwide Operations until it was acquired by Oceanic Drives Corporation in May 2016. Mr. Chen holds a BS and MS in Mechanical Engineering and an MBA, each from the East Coast Institute of Technology. Senior Vice President, Worldwide SalesMr. Bokan, 62, joined us in 1996 and has served in various leadership positions since that time. Mr. Bokan was named Senior Vice President, Worldwide Sales in October 2018. Mr. Bokan holds a BS in Business Administration from Mountain State University. Executive Vice President, Technology & ProductsDr. Roe, 57, joined us in February 1995 and has served in various leadership positions since that time. Dr. Roe was named Executive Vice President, Technology Development in June 2017 and named Executive Vice President, Technology & Products in September 2019. Dr. Roe holds a PhD in Electrical Engineering and an MS in Physics from Iowa State University. He completed his undergraduate degree at Plains College.

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President, Chief Executive Officer, and DirectorMr. James Harrington, 65, joined us in May 2017 as our President, Chief Executive Officer, and Director. Mr. James Harrington co-founded and led StorEdge Technologies Corporation as a start-up in 1988 until its eventual sale in May 2016, serving as its President and Chief Executive Officer from January 2011 to May 2016, and as a member of its Board of Directors from July 2010 to May 2016. Mr. James Harrington served as a member of the Board of Directors for Nexus Semiconductors, Inc. from July 2009 until July 2018 and for Oceanic Drives Corp. from May 2016 to February 2017 and has served since March 2021 as a member of the Board of Directors of TechSupply Co.. Mr. James Harrington holds a BS and an MS in Electrical Engineering and Computer Science from the West Coast Tech University and is a graduate of the West Coast Tech Graduate School of Business Executive Program. Executive Vice President and Chief Financial OfficerMr. Michael Sterling, 55, joined us in April 2022 as Executive Vice President and Chief Financial Officer. From June 2016 to April 2022, Mr. Michael Sterling served as the Chief Financial Officer of Radiowave Inc. Prior to Radiowave, Mr. Michael Sterling served as Executive Vice President and Chief Financial Officer of Apex Auto Parts PLC, and prior to Apex Auto, held executive roles at Aerogas Inc. and Apex Substrate Materials, Inc. Mr. Michael Sterling currently serves on the Board of Directors of Continental Specialty Corp. Mr. Michael Sterling is a veteran of the U.S. Marine Corps and holds an MBA from Northeastern Business School and BS in Business from Great Lakes University. Executive Vice President and Chief Business OfficerMr. Lipton, 54, joined us in June 2017 as our Executive Vice President and Chief Business Officer. From April 2010 to May 2016, Mr. Lipton served in various roles at StorEdge Technologies Corp., including Executive Vice President, Chief Strategy Officer, and General Manager, Enterprise Solutions until it was acquired by Oceanic Drives in May 2016. Mr. Lipton currently serves on the Board of Directors of Silicon Laboratories, Inc. Mr. Lipton holds a B. Tech. in Electrical Engineering from the South Asian Technical Institute, Nashik, India and an MS in Electrical Engineering from West Coast Tech University.

There are no family relationships between any of our directors or executive officers.

Available Information

Our executive offices are located at 5000 Oceanic Way, Reno, Nevada 83750 and our telephone number is (775) 555-0133. Information about us is available on our website, www.northwindmemory.com. Also available on our website are our Corporate Governance Guidelines, Governance and Sustainability Committee Charter, Compensation Committee Charter, Audit Committee Charter, Finance Committee Charter, Security Committee Charter, and Code of Business Conduct and Ethics. We intend to satisfy the disclosure requirement under Item 5.05 of Current Event Disclosure regarding any amendments to, or waivers from, our Code of Business Conduct and Ethics by posting such information on our website within four business days of the amendment or waiver. Copies of these documents are available to shareholders upon request. Information contained or referenced on our website is not incorporated by reference and does not form a part of this Annual Report on Annual Statutory Report. Investors and others should note that we announce material financial information about our business and products

through a variety of means, including our investor relations website (investors.northwindmemory.com), filings with the National Financial Oversight Commission (“Financial Oversight Commission”), press releases, public conference calls, blog posts (northwindmemory.com/about/blog), and webcasts. We use these channels to achieve broad, non-exclusionary distribution of information to the public and for complying with our disclosure obligations under Regulation FD. Therefore, we encourage investors, the media, and others interested in our company to review the information we post on such channels. Our filings are available free of charge on our website as soon as reasonably practicable after they are electronically filed with, or furnished to, the Financial Oversight Commission, including our annual and quarterly reports on Forms 10-K and 10-Q and current reports on Current Event Disclosure, our proxy statements, and any amendments to those reports or statements. The Financial Oversight Commission’s website, www.financial-oversight-commission.gov, contains reports, proxy and information statements, and other information regarding issuers that file electronically with the Financial Oversight Commission. The content on any website referred to in this Annual Statutory Report is not incorporated by reference in this Annual Statutory Report unless expressly noted.

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Item 1A: RISK FACTORS

ITEM 1A. RISK FACTORS

In addition to the factors discussed elsewhere in this Annual Statutory Report, this section discusses important factors which could cause actual results or events to differ materially from those contained in any forward-looking statements made by us. The order of presentation is not necessarily indicative of the level of risk that each factor poses to us. Any of these factors could have a material adverse effect on our business, results of operations, financial condition, or stock price. Our operations could also be affected by other factors that are presently unknown to us or not considered significant.

Risk Factor Summary

Risks Related to Our Business, Operations, and Industry

- volatility in average selling prices of our products;
- a range of factors that may adversely affect our gross margins;
- our international operations, including geopolitical risks;
- the highly competitive nature of our industry;
- our ability to develop and produce new and competitive memory and storage technologies and products;
- realizing expected returns from capacity expansions;
- achieving or maintaining certain performance or other obligations associated with incentives from various governments;
- availability and quality of materials, supplies, and capital equipment and dependency on third-party service providers;
- a downturn in regional or worldwide economies;
- disruptions to our manufacturing process from operational issues, natural disasters, or other events;
- dependency on a select number of key customers, including international customers;
- products that fail to meet specifications, are defective, or are incompatible with end uses;
- breaches of our security systems or products, or those of our customers, suppliers, or business partners;
- attracting, retaining, and motivating highly skilled employees;

- responsible sourcing requirements and related regulations;
- environmental, social, and governance considerations;
- acquisitions and/or alliances; and
- restructure plans may not realize expected savings or other benefits.

Risks Related to Intellectual Property and Litigation

•protecting our intellectual property and retaining key employees who are knowledgeable of and develop our intellectual property;

- legal proceedings and claims; and
- claims

that our products or manufacturing processes infringe or otherwise violate the intellectual property rights of others or failure to obtain or renew license agreements covering such intellectual property.

Risks Related to Laws and Regulations

- impacts of government actions and compliance with tariffs, trade restrictions, and/or trade regulations;
- tax expense and tax laws in key jurisdictions; and
- compliance with laws, regulations, or industry standards, including environmental considerations.

Risks Related to Capitalization and Financial Markets

•our ability to generate sufficient cash flows or obtain access to external financing;

- our debt obligations;
- changes in foreign currency exchange rates;
- counterparty default risk;
- volatility in the trading price of our common stock; and
- fluctuations in the amount and frequency of our common stock repurchases and payment of cash dividends and resulting impacts.

Risks Related to Our Business, Operations, and Industry

Volatility in average selling prices for our semiconductor memory and storage products may adversely affect our business.

We have experienced significant volatility in our average selling prices and may continue to experience such volatility in the future. For example, average selling prices for DRAM declined in the high-40s percent range and NAND declined in the low-50s percent range for 2023 as compared to 2022. Since 2017, annual percentage changes in DRAM average selling prices have ranged from approximately plus 35% to a minus high-40s percent range. Since 2017, annual percentage changes in NAND average selling prices have ranged from nearly flat to a minus low-50s percent range. In current and recent periods, average selling prices for our products have been below our manufacturing costs and we may experience such circumstances in the future. Average selling prices for our products that decline faster than our costs have recently had an adverse effect on our business and results of operations, and in future periods could have a material adverse effect on our business, results of operations, or financial condition.

Our gross margins may be adversely affected by a range of factors.

Our gross margins are dependent, in part, upon continuing decreases in per gigabit manufacturing costs achieved through improvements in our manufacturing processes and product designs. Factors that may limit our ability to reduce our per gigabit manufacturing costs at sufficient levels to prevent deterioration of or improve gross margins include, but are not limited to:

- strategic product diversification decisions affecting product mix;
- increasing complexity of manufacturing processes;
- difficulties in transitioning to smaller line-width process technologies or additional 3D memory layers or NAND cell levels;
- process complexity including number of mask layers and fabrication steps;
- manufacturing yield;
- technological barriers;
- changes in process technologies;
- new products that may require relatively larger die sizes;
- start-up or other costs associated with capacity expansions;
- higher costs of goods and services due to inflationary pressures or market conditions; and
- higher

manufacturing costs per gigabit due to fabrication facility underutilization, lower wafer output, and insufficient volume to run new technology nodes to achieve cost optimization. Many factors may result in a reduction of our output or a delay in ramping production, which could lead to underutilization of our production assets. These factors may include, among others, a weak demand environment, industry oversupply, inventory surpluses, difficulties in ramping emerging technologies, supply chain disruptions, and delays from equipment suppliers. See “Part II – Item 7. Management’s Discussion and Analysis of Financial Condition and Results of Operations – Overview – Industry Conditions” for information regarding our current underutilization. A significant portion of our manufacturing costs are fixed and do not vary proportionally with changes in production output. As a result, lower utilization, lower wafer output, and corresponding increases in our per gigabit manufacturing costs have resulted in higher inventory carrying costs, and have had, and may continue to have, an adverse effect on our gross margins, business, results of operations, or financial condition.

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We have a broad portfolio of products to address our customers’ needs, which span multiple market segments and are subject to rapid technological changes. Our manufacturing costs on a per gigabit basis vary across our portfolio as they are largely influenced by the technology node in which the solution was developed. We strive to balance our demand and supply for each technology node, but the dynamics of our markets and our customers can create periods of imbalance, which can lead us to carry elevated inventory levels. Consequently, we may incur charges in connection with obsolete or excess inventories, or we may not fully recover our costs, which would reduce our gross margins. For example, in 2023, we recorded aggregate charges of \$2.38 billion to write down the carrying value of our inventories to their estimated net realizable value. In addition, due to the customized nature of certain products we manufacture, we may be unable to sell certain finished goods inventories to alternative customers or manufacture in-process inventory to different specifications, which may result in excess and obsolescence charges in future periods. In addition, if we are unable to supply products that meet customer design and performance specifications, we may be required to sell such products at lower average selling prices, which may reduce our gross margins. Our gross margins may also be impacted by shifts in product mix, driven by our strategy to optimize our portfolio to best respond to changing market dynamics. Our inability to prevent deterioration of or improve gross margins could have a material adverse effect on our business, results of operations, or financial condition.

We face geopolitical and other risks associated with our international operations that could materially adversely affect our business, results of operations, or financial condition.

In addition to our U.S. operations, a substantial portion of our operations are conducted in Taiwan, Singapore, Japan, Malaysia, China, and India, and many of our customers, suppliers, and vendors also operate internationally. In 2023, nearly half of our revenue was from sales to customers who have headquarters located outside the United States, while over 80% of our revenue in 2023 was from

products shipped to customer locations outside the United States. Our international operations are subject to a number of risks, including:

- restrictions on sales of goods or services to one or more of our significant foreign customers;
- export

and import duties, changes to import and export regulations, customs regulations and processes, and restrictions on the transfer of funds, including currency controls in China, which could negatively affect the amount and timing of payments from certain of our customers and, as a result, our cash flows;

- compliance

with U.S. and international laws involving international operations, including the Foreign Corrupt Practices Act of 1977, as amended, sanctions and anti-corruption laws, export and import laws, and similar rules and regulations;

- theft of intellectual property;
- political and economic instability, including instability resulting from domestic and international conflicts;
- government

actions or civil unrest preventing the flow of products and materials, including delays in shipping and obtaining products and materials, cancellation of orders, or loss or damage of products;

- problems with the transportation or delivery of products and materials;
- issues arising from cultural or language differences and labor unrest;
- longer payment cycles and greater difficulty in collecting accounts receivable;
- compliance with trade, technical standards, and other laws in a variety of jurisdictions;
- contractual and regulatory limitations on the ability to maintain flexibility with staffing levels;
- disruptions to manufacturing or R&D activities as a result of actions imposed by foreign governments;
- changes in economic policies of foreign governments;
- difficulties in staffing and managing international operations; and
- public health issues. If

we or our customers, suppliers, or vendors are impacted by any of these risks, it could have a material adverse effect on our business, results of operations, or financial condition. Following the May 26, 2023 decision of its cybersecurity review of our products sold in China, the NCRB determined that critical information infrastructure operators in China may not purchase Northwind products, impacting our revenue with companies headquartered in mainland China and Hong Kong, including direct sales as well as indirect sales through distributors. Some revenue with customers headquartered outside of China has also been impacted. As we try to mitigate possible impacts due to the NCRB decision, revenue may come at lower prices or gross margins due to product or customer mix changes, which may impact our business results. Further actions by the Chinese government could impact additional revenue inside or outside China, or our operations in China, or our ability to ship products to our customers, any of which could have a material adverse effect on our business, results of operations, or financial condition. Political, economic, or other actions may adversely affect our operations in Taiwan. A majority of our DRAM production output in 2023 was from our fabrication facilities in Taiwan and any loss of output could have a material adverse effect on us. Any political, economic, or other actions may also adversely affect our customers and the technology industry supply chain, for which Taiwan is a central hub, and as a result, could have a material adverse impact on us. In addition, the U.S. government has in the past restricted American firms from selling products and software to certain of our customers and may in the future impose similar restrictions on one or more of our significant customers. These restrictions may not prohibit our competitors from selling similar products to our customers, which may result in our loss of sales and market share. Even as such restrictions are lifted, financial or other penalties or continuing

export restrictions imposed with respect to our customers could have a continuing negative impact on our future revenue and results of operations, and we may not be able to recover any customers or market share we lose, or make such recoveries at acceptable average selling prices, while complying with such restrictions.

The semiconductor memory and storage markets are highly competitive.

We face intense competition in the semiconductor memory and storage markets from a number of companies, including Memorytek Holdings Corporation; Orion Electronics Co., Ltd.; Peak Semiconductor Inc.; and Oceanic Drives Corporation. Our competitors may use aggressive pricing to obtain market share. Some of our competitors are large corporations or conglomerates that may have a larger market share and greater resources to invest in technology, capitalize on growth opportunities, and withstand downturns in the semiconductor markets in which we compete. Consolidation of industry competitors could put us at a competitive disadvantage as our competitors may benefit from increased manufacturing scale and a stronger product portfolio. We operate in different jurisdictions than our competitors and may be impacted by unfavorable changes in currency exchange rates. In addition, some governments may provide, or have provided and may continue to provide, significant assistance, financial or otherwise, to some of our competitors or to new entrants and may intervene in support of national industries and/or competitors. In particular, we face the threat of increasing competition as a result of significant investment in the semiconductor industry by the Chinese government and various state-owned or affiliated entities, in companies such as Eastern Memory Technologies Co., Ltd. (“EMC”) and Orient Chips Inc. (“OCI”). In addition, the NCRB’s decision that critical information infrastructure operators in China may not purchase Northwind products had an impact on our ability to compete effectively in China and elsewhere. We and our competitors generally seek to increase wafer output, improve yields, and reduce die size, which could result in significant increases in worldwide supply and downward pressure on prices. Increases in worldwide supply of semiconductor memory and storage also result from fabrication capacity expansions, either by way of new facilities, increased capacity utilization, or reallocation of other semiconductor production to semiconductor memory and storage production. Our competitors may increase capital expenditures resulting in future increases in worldwide supply. We, and some of our competitors, have plans to ramp, or are constructing or ramping, production at new fabrication facilities. Increases in worldwide supply of semiconductor memory and storage, if not accompanied by commensurate increases in demand, could lead to declines in average selling prices for our products and could materially adversely affect our business, results of operations, or financial condition. If competitors are more successful at developing or implementing new product or process technology, their products could have cost or performance advantage. The competitive nature of our industry could have a material adverse effect on our business, results of operations, or financial condition.

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Our future success depends on our ability to develop and produce new and competitive memory and storage technologies and products.

Our key semiconductor memory and storage technologies face technological barriers to continue to meet long-term customer needs. These barriers include potential limitations on stacking additional 3D memory layers, increasing bits per cell (i.e., cell levels), meeting higher density requirements, improving power consumption and reliability, and delivering advanced features and higher performance. We may face technological barriers to continue to shrink our products at our current or historical rate, which has generally reduced per gigabit cost. We have invested and expect to continue to invest in R&D for new and existing products and process technologies, such as EUV lithography, to continue to deliver advanced product requirements. Such new technologies can add complexity and risk to our schedule and may affect our costs and production output. We may be unable to recover our investment in R&D or otherwise realize the economic benefits of reducing die size or increasing memory and storage densities. Our competitors are working to develop new memory and storage technologies that may offer

performance and/or cost advantage to existing technologies and render existing technologies obsolete. Accordingly, our future success may depend on our ability to develop and produce viable and competitive new memory and storage technologies. We are developing new products, including system-level memory and storage products and solutions, which complement our traditional products or leverage their underlying design or process technology. We have invested and expect to continue to invest in new semiconductor product and system-level solution development. We are increasingly differentiating our products and solutions to meet the specific demands of our customers, which increases our reliance on our customers' ability to accurately forecast the needs and preferences of their customers. As a result, our product demand forecasts may be impacted significantly by the strategic actions of our customers. In addition, our ability to successfully introduce new products often requires us to make product specification decisions multiple years in advance of when new products enter the market. It is important that we deliver products in a timely manner with increasingly advanced performance characteristics at the time our customers are designing and evaluating samples for their products. If we do not meet their product design schedules, our customers may exclude us from further consideration as a supplier for those products. The process to develop new products requires us to demonstrate advanced functionality, performance, and reliability, often well in advance of a planned ramp of production, in order to secure design wins with our customers. Many factors may negatively impact our ability to meet anticipated timelines and/or expected or required quality standards with respect to the development of certain of our products. In addition, some of our components have long lead-times, requiring us to place orders up to a year in advance of anticipated demand. Such long lead-times increase the risk of excess inventory or loss of sales in the event our forecasts vary substantially from actual demand. There can be no assurance of the following:

- we will be successful in developing competitive new semiconductor memory and storage technologies and products;
- we will be able to cost-effectively manufacture new products;
- we will be able to successfully market these technologies;
- margins generated from sales of these products will allow us to recover costs of development efforts;
- we

will be able to establish or maintain key relationships with customers, or that we will not be prohibited from working with certain customers, for specific chip set or design requirements;

- we will accurately predict and design products that meet our customers' specifications; or
- we will be able to introduce new products into the market and qualify them with our customers on a timely basis. Unsuccessful

efforts to develop new memory and storage technologies and products could have a material adverse effect on our business, results of operations, or financial condition.

We may not be able to achieve expected returns from capacity expansions.

We have announced our intent to expand our production capacity and/or make capital investments in the United States and in other regions where we operate. These expansions involve several risks including the following:

- capital

expenditure requirements for capacity expansions during periods of relatively low free cash flow generation, resulting from challenging memory and storage industry conditions;

- availability of necessary funding, which may include external sources;
- ability

to realize expected grants, investment tax credits, and other government incentives, including through the U.S. CHIPS and Science Act of 2022 ("CHIPS Act") and other national, international, state, and local

grants;

- potential changes in laws or provisions of grants, investment tax credits, and other government incentives;
- potential restrictions on expanding in certain geographies;
- availability of equipment and construction materials;
- ability to complete construction as scheduled and within budget;
- availability of the necessary workforce;
- ability to timely ramp production in a cost-effective manner;
- increases to our cost structure until new production is ramped to adequate scale; and
- sufficient customer demand to utilize our increased capacity. We

invest our capital in areas that we believe best align with our business strategy and optimize future returns. Investments in capital expenditures may not generate expected returns or cash flows. Significant judgment is required to determine which capital investments will result in optimal returns, and we could invest in projects that are ultimately less profitable than those projects we do not select. Delays in completion and ramping of new production facilities, or failure to optimize our investment choices, could significantly impact our ability to realize expected returns on our capital expenditures. Any of the above factors could have a material adverse effect on our business, results of operations, or financial condition.

Our incentives from various governments are conditional upon achieving or maintaining certain performance or other obligations and are subject to reduction, termination, clawback, or could impose certain limitations on our business.

We have received, and may in the future continue to receive, benefits and incentives from national, state, and local governments in various regions of the world designed to encourage us to establish, maintain, or increase investment, workforce, or production in those regions. These incentives may take various forms, including grants, loan subsidies, and tax arrangements, and typically require us to achieve or maintain certain levels of investment, capital spending, employment, technology deployment, or research and development activities to qualify for such incentives or could restrict us from undertaking certain activities. We may be unable to obtain significant future incentives to continue to fund a portion of our capital expenditures and operating costs, without which our cost structure would be adversely impacted. We also cannot guarantee that we will successfully achieve performance or other obligations required to qualify for these incentives or that the granting agencies will provide such funding. These incentive arrangements typically provide the granting agencies with rights to audit our compliance with their terms and obligations. Such audits could result in modifications to, or termination of, the applicable incentive program. The incentives we receive could be subject to reduction, termination, or clawback, and any decrease or clawback of government incentives could have a material adverse effect on our business, results of operations, or financial condition.

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Our business, results of operations, or financial condition could be adversely affected by the availability and quality of materials, supplies, and capital equipment, or dependency on third-party service providers.

Our supply chain and operations are dependent on the availability of materials that meet exacting standards and the use of third parties to provide us with components and services. We generally have multiple sources of supply for our materials and services. However, only a limited number of suppliers are capable of delivering certain materials, components, and services that meet our standards and, in some cases, materials, components, or services are provided by a single or sole source, and we may be unable to qualify new suppliers on a timely basis. The availability of materials or components such as chemicals, silicon wafers, gases, photoresist, controllers, substrates, lead frames, printed c boards, targets, and reticle glass blanks is impacted by various factors. These factors could include a shortage of

raw materials or a disruption in the processing or purification of those raw materials into finished goods. Shortages or increases in lead times have occurred in the past, are currently occurring with respect to some materials and components, and may occur from time to time in the future. Constraints within our supply chain for certain materials and integrated circuit components could limit our bit shipments, which could have a material adverse effect on our business, results of operations, or financial condition. Our manufacturing processes are also dependent on our relationships with third-party manufacturers of controllers, analog integrated circuits, and other components used in some of our products and with outsourced semiconductor foundries, assembly and test providers, contract manufacturers, logistics carriers, and other service providers, including providers of electricity and other utilities. Although we have certain long-term contracts with some of our suppliers, many of these contracts do not provide for long-term capacity or pricing commitments. To the extent we do not have firm commitments from our third-party suppliers over a specific time period or for any specific capacity, quantity, and/or pricing, our suppliers may allocate capacity to their other customers and capacity and/or materials may not be available when needed or at reasonable prices. Inflationary pressures have increased, and may continue to increase costs for materials, supplies, and services. Regardless of contract structure, large swings in demand may exceed our contracted supply and/or our suppliers' capacity to meet those demand changes resulting in a shortage of parts, materials, or capacity needed to manufacture our products. In addition, if any of our suppliers was to cease operations or become insolvent, this could impact their ability to provide us with necessary supplies, and we may not be able to obtain the needed supply in a timely way or at all from other providers. Certain materials are primarily available in a limited number of countries, including rare earth elements, minerals, and metals. Trade disputes, geopolitical tensions, economic circumstances, political conditions, or public health issues may limit our ability to obtain such materials. Although these rare earth and other materials are generally available from multiple suppliers, China is the predominant producer of certain of these materials. If China were to restrict or stop exporting these materials, our suppliers' ability to obtain such supply may be constrained and we may be unable to obtain sufficient quantities, or obtain supply in a timely manner, or at a commercially reasonable cost. Constrained supply of rare earth elements, minerals, and metals may restrict our ability to manufacture certain of our products and make it difficult or impossible to compete with other semiconductor memory and storage manufacturers who are able to obtain sufficient quantities of these materials from China. We and/or our suppliers and service providers could be affected by regional conflicts, civil unrest, labor disruptions, sanctions, tariffs, embargoes, or other trade restrictions, as well as laws and regulations enacted in response to concerns regarding climate change, conflict minerals, responsible sourcing practices, public health crises, or other matters, which could limit the supply of our materials and/or increase the cost. Environmental regulations could limit our ability to procure or use certain chemicals or materials in our operations or products. In addition, disruptions in transportation lines could delay our receipt of materials. Our ability to procure components to repair equipment essential for our manufacturing processes could also be negatively impacted by various restrictions or disruptions in supply chains, among other items. The disruption of our supply of materials, components, or services, or the extension of our lead times could have a material adverse effect on our business, results of operations, or financial condition. Our operations are dependent on our ability to procure advanced semiconductor manufacturing equipment that enables the transition to lower cost manufacturing processes. For certain key types of equipment, including photolithography tools, we are sometimes dependent on a single supplier. From time to time, we have experienced difficulties in obtaining some equipment on a timely basis due to suppliers' limited capacity. Our inability to obtain equipment on a timely basis could adversely affect our ability to transition to next generation manufacturing processes and reduce our costs. Delays in obtaining equipment could also impede our ability to ramp production and could increase our overall costs of a ramp. Our inability to obtain advanced semiconductor manufacturing equipment in a timely manner could have a material adverse effect on our business, results of operations, or financial condition. Our construction projects to expand production and R&D capacity are highly dependent on available sources of labor, materials, equipment, and services. Increasing demand, supply constraints, inflation, and other market conditions could result in increasing shortages and higher costs for these items. Difficulties in obtaining these resources could result in significant delays in completion of our construction projects and cost increases, which could have a

material adverse effect on our business, results of operations, or financial condition. Our inability to source materials, supplies, capital equipment, or third-party services could affect our overall production output and our ability to fulfill customer demand. Significant or prolonged shortages of our products could halt customer manufacturing and damage our relationships with these customers. Any damage to our customer relationships as a result of a shortage of our products could have a material adverse effect on our business, results of operations, or financial condition. Similarly, if our customers experience disruptions to their supplies, materials, components, or services, or the extension of their lead times, they may reduce, cancel, or alter the timing of their purchases with us, which could have a material adverse effect on our business, results of operations, or financial condition.

Downturns in regional or worldwide economies may harm our business.

Downturns in regional or worldwide economies, due to inflation, geopolitics, major central bank policy actions including interest rate increases, public health crises, or other factors, have harmed our business in the past and current and future downturns could also adversely affect our business. Adverse economic conditions affect demand for devices that incorporate our products, such as personal computers, smartphones, automobiles, and servers. Reduced demand for these or other products could result in significant decreases in our average selling prices and product sales. In addition, to the extent our customers or distributors have elevated inventory levels or are impacted by a deterioration in credit markets, we may experience a decrease in short-term and/or long-term demand resulting in industry oversupply and declines in pricing for our products. A deterioration of conditions in regional or worldwide credit markets could limit our ability to obtain external financing to fund our operations and capital expenditures. In addition, we may experience losses on our holdings of cash and investments due to failures of financial institutions and other parties. Difficult economic conditions may also result in a higher rate of losses on our accounts receivable due to credit defaults. As a result, downturns in regional or worldwide economies could have a material adverse effect on our business, results of operations, or financial condition.

If our manufacturing process is disrupted by operational issues, natural disasters, or other events, our business, results of operations, or financial condition could be materially adversely affected.

We and our subcontractors manufacture products using highly complex processes that require technologically advanced equipment and continuous modification to improve yields and performance. Difficulties in the manufacturing process or the effects from a shift in product mix can reduce yields or disrupt production and may increase our per gigabit manufacturing costs. We and our subcontractors maintain operations and continuously implement new product and process technology at manufacturing facilities, which are widely dispersed in multiple locations in several countries including the United States, Singapore, Taiwan, Japan, Malaysia, and China. As a result of the necessary interdependence within our network of manufacturing facilities, an operational disruption at one of our or a subcontractor's facilities may have a disproportionate impact on our ability to produce many of our products.

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From time to time, there have been disruptions in our manufacturing operations as a result of power outages, improperly functioning equipment, disruptions in supply of raw materials or components, or equipment failures. We have manufacturing and other operations in locations subject to natural occurrences and possible climate changes, such as severe and variable weather and geological events resulting in increased costs, or disruptions to our manufacturing operations or those of our suppliers or customers. In addition, climate change may pose physical risks to our manufacturing facilities or our suppliers' facilities, including increased extreme weather events that could result in supply delays or disruptions. Other events, including political or public health crises, such as an outbreak of contagious diseases, may also affect our production capabilities or that of our suppliers, including as a result of quarantines, closures of production facilities, lack of supplies, or delays caused by restrictions on travel or shipping. Events of the types noted above have occurred from time to time and may occur in the

future. As a result, in addition to disruptions to operations, our insurance premiums may increase or we may not be able to fully recover any sustained losses through insurance. If production is disrupted for any reason, manufacturing yields may be adversely affected, or we may be unable to meet our customers' requirements and they may purchase products from other suppliers. This could result in a significant increase in manufacturing costs, loss of revenue, or damage to customer relationships, any of which could have a material adverse effect on our business, results of operations, or financial condition.

A significant portion of our revenue is concentrated with a select number of customers.

In each of the last three years, approximately one-half of our total revenue was from our top ten customers. A disruption in our relationship with any of these customers could adversely affect our business. We could experience fluctuations in our customer base or the mix of revenue by customer as markets and strategies evolve. Our customers' demand for our products may fluctuate due to factors beyond our control. In addition, any consolidation of our customers could reduce the number of customers to whom our products may be sold. Our inability to meet our customers' requirements or to qualify our products with them could adversely impact our revenue. A meaningful change in the inventory strategy of our customers could impact our industry bit demand growth outlook. The loss of, or restrictions on our ability to sell to, one or more of our major customers, or any significant reduction in orders from, or a shift in product mix by, customers could have a material adverse effect on our business, results of operations, or financial condition.

Increases in sales of system solutions may increase our dependency upon specific customers and our costs to develop, qualify, and manufacture our system solutions.

Our development of system-level memory and storage products is dependent, in part, upon successfully identifying and meeting our customers' specifications for those products. Developing and manufacturing system-level products with specifications unique to a customer increases our reliance upon that customer for purchasing our products at sufficient volumes and prices in a timely manner. Even if our products meet customer specifications, our sales of system-level solutions are dependent upon our customers choosing our products over those of our competitors and purchasing our products at sufficient volumes and prices. Our competitors' products may be less costly, provide better performance, or include additional features when compared to our products. Our long-term ability to sell system-level memory and storage products is reliant upon our customers' ability to create, market, and sell their products containing our system-level solutions at sufficient volumes and prices in a timely manner. If we fail to successfully develop and market system-level products, our business, results of operations, or financial condition may be materially adversely affected. Manufacturing system-level solutions, such as SSDs, managed NAND, and HBM, typically results in higher per-unit manufacturing costs as compared to other products. Even if we are successful in selling system-level solutions to our customers in sufficient volume, we may be unable to generate sufficient profit if our per-unit manufacturing costs are not offset by higher per-unit selling prices. Manufacturing system-level solutions to customer specifications requires a longer development cycle, as compared to discrete products, to design, test, and qualify, which may increase our costs. Some of our system solutions are increasingly dependent on sophisticated firmware that may require significant customization to meet customer specifications, which increases our costs and time to market. Additionally, we may need to update our controller and hardware design as well as our firmware or develop new firmware as a result of new product introductions or changes in customer specifications and/or industry standards, which increases our costs. System complexities and extended warranties for system-level products could also increase our warranty costs. Our failure to cost-effectively manufacture system-level solutions and/or controller, hardware design, and firmware in a timely manner may result in reduced demand for our system-level products and could have a material adverse effect on our business, results of operations, or financial condition.

Products that fail to meet specifications, are defective, or are otherwise incompatible with end uses could impose significant costs on us.

Products that do not meet specifications or that contain, or are perceived by our customers to contain, defects or that are otherwise incompatible with end uses could impose significant costs on us or otherwise materially adversely affect our business, results of operations, or financial condition. From time to time, we experience problems with nonconforming, defective, or incompatible products after we have shipped such products. In recent periods, we have further diversified and expanded our product offerings, which could potentially increase the chance that one or more of our products could fail to meet specifications in a particular application. Our products and solutions may be deemed fully or partially responsible for functionality in our customers' products and may result in sharing or shifting of product or financial liability from our customers to us for costs incurred by the end user as a result of our customers' products failing to perform as specified. In addition, if our products and solutions perform critical functions in our customers' products or are used in high-risk consumer end products, such as autonomous driver assistance programs, home and enterprise security, smoke and noxious gas detectors, medical monitoring equipment, or wearables for child and elderly safety, our potential liability may increase. We could be adversely affected in several ways, including the following:

- we

may be required or agree to compensate customers for costs incurred or damages caused by defective or incompatible products and to replace products;

- we

could incur a decrease in revenue or adjustment to pricing commensurate with the reimbursement of such costs or alleged damages; and

- we

may encounter adverse publicity, which could cause a decrease in sales of our products or harm our reputation or relationships with existing or potential customers. Any of the foregoing items could have a material adverse effect on our business, results of operations, or financial condition.

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Breaches of our security systems or products, or those of our customers, suppliers, or business partners, could expose us to losses.

We maintain a system of controls over the physical security of our facilities. We also manage and store various proprietary information and sensitive or confidential data relating to our operations. In addition, we process, store, and transmit large amounts of data relating to our customers and employees, including sensitive personal information. Unauthorized persons, employees, former employees, nation states, or other parties may gain access to our facilities or technology infrastructure and systems to steal trade secrets or other proprietary information, compromise confidential information, create system disruptions, or cause shutdowns. This risk is exacerbated as competitors for talent, particularly engineering talent, attempt to hire our employees. Through cyberattacks on technology infrastructure and systems, unauthorized parties may obtain access to computer systems, networks, and data, including cloud-based platforms. The technology infrastructure and systems of our suppliers, vendors, service providers, cloud solution providers, and partners have in the past experienced, and may in the future experience, such attacks, which could impact our operations. Cyberattacks can include ransomware, computer denial-of-service attacks, worms, supply chain attacks, social engineering, open source vulnerabilities, and other malicious software programs or other attacks, including those using techniques that change frequently or may be disguised or difficult to detect, or designed to remain dormant until a triggering event, impersonation of authorized users, and efforts to discover and exploit any design flaws, "bugs," security vulnerabilities, as well as intentional or unintentional acts by employees or other insiders with access privileges. Additionally, some actors are using artificial intelligence technology to launch more automated, targeted and coordinated attacks. Globally, cyberattacks are increasing in number and the attackers are increasingly organized and well-financed, or supported by state actors, and are developing increasingly sophisticated systems to not only attack, but also to evade detection. In addition,

geopolitical tensions or conflicts may create a heightened risk of cyberattacks. Breaches of our physical security, attacks on our technology infrastructure and systems, or breaches or attacks on our customers, suppliers, or business partners who have confidential or sensitive information regarding us and our customers and suppliers, could result in significant losses and damage our reputation with customers and suppliers and may expose us to litigation if the confidential information of our customers, suppliers, or employees is compromised. Our products are also targets for cyberattacks, including those products utilized in cloud-based environments. While some of our products contain encryption or security algorithms to protect third-party content or user-generated data stored on our products, these products could still be hacked or the encryption schemes could be compromised, breached, or circumvented by motivated and sophisticated attackers. Further, our products contain sophisticated hardware and firmware and applications that may contain security vulnerabilities or defects in design or manufacture, including “bugs” and other problems that could interfere with the intended operation of our products. To the extent our products are hacked, or the encryption schemes are compromised or breached, this could harm our business by requiring us to employ additional resources to fix the errors or defects, exposing us to litigation, claims, and harm to our reputation. Any of the foregoing security risks could have a material adverse effect on our business, results of operations, or financial condition.

We must attract, retain, and motivate highly skilled employees.

To remain competitive, we must attract, retain, and motivate executives and other highly skilled, diverse employees, as well as effectively manage succession for key employees. Competition for experienced employees in our industry can be intense. Hiring and retaining qualified executives and other employees is critical to our business. If our total compensation programs, employment benefits, and workplace culture are not viewed as competitive and inclusive, our ability to attract, retain, and motivate employees could be compromised. At times, we experience higher levels of attrition, increasing compensation costs, and more intense competition for talent across our industry. To the extent we experience significant attrition and are unable to timely replace employees, we could experience a loss of critical skills and reduced employee morale, potentially resulting in business disruptions or increased expenses to address any disruptions. Additionally, changes to immigration policies in the countries in which we operate, as well as restrictions on travel due to public health crises or other causes, may limit our ability to hire and/or retain talent in, or transfer talent to, specific locations. Our inability to attract, retain, and motivate executives and other employees or effectively manage succession of key roles may inhibit our ability to maintain or expand our business operations.

Compliance with responsible sourcing requirements and any related regulations could increase our operating costs, or limit the supply and increase the cost of certain materials, supplies, and services, and if we fail to comply, customers may reduce purchases from us or disqualify us as a supplier.

We and many of our customers have adopted responsible sourcing programs that require us to meet certain environmental, social and governance criteria, and to periodically report on our performance against these requirements, including that we source the materials, supplies, and services we use and incorporate into the products we sell as prescribed by these programs. Many customer programs require us to remove a supplier within a prescribed period if such supplier ceases to comply with prescribed criteria, and our supply chain may at any time contain suppliers at risk of being removed due to non-compliance with responsible sourcing requirements. Some of our customers may elect to disqualify us as a supplier (resulting in a permanent or temporary loss of sales to such customer) or reduce purchases from us if we are unable to verify that our performance or products (including the underlying supply chain) meet the specifications of our customers’ responsible sourcing programs on a continuous basis. Meeting responsible sourcing requirements may increase operating requirements and costs or limit the sourcing and availability of some of the materials, supplies, and services we use, particularly when the availability of such materials, supplies, and services is concentrated to a limited number of suppliers. From time to time, we remove suppliers or require our suppliers to remove suppliers from their supply chains based on our responsible sourcing requirements or customer requirements, and we or our suppliers may be unable to replace such removed suppliers in a timely or cost-effective manner. Any

inability to replace removed suppliers in a timely or cost effective manner may affect our ability and/or the cost to obtain sufficient quantities of materials, supplies, and services necessary for the manufacture of our products. Our inability to replace suppliers we have removed in a timely or cost-effective manner or comply with customers' responsible sourcing requirements or with any related regulations could have a material adverse effect on our business, results of operations, or financial condition.

Failure to meet environmental, social, and governance expectations or standards or achieve our related goals could adversely affect our business, results of operations, financial condition, or stock price.

In recent years, there has been an increased focus from stakeholders on environmental, social, and governance matters, including greenhouse gas emissions and climate-related risks, sustainability, renewable energy, water stewardship, waste management, diversity, equality and inclusion, responsible sourcing and supply chain, human rights, and social responsibility. Given our commitment to relevant social and environmental issues as it relates to our business, we actively manage these issues and have established and publicly announced certain goals, commitments, and targets which we may refine or even expand further in the future. These goals, commitments, and targets reflect our current plans and aspirations and are not guarantees that we will be able to achieve them. Achieving these goals may entail significant costs, for example we have entered into several virtual power purchase agreements to obtain renewable energy credits at a cost that will vary based on future prices for electrical power. Evolving stakeholder expectations and our efforts to manage these issues, report on them, and accomplish our goals present numerous operational, regulatory, reputational, financial, legal, and other risks, any of which could have a material adverse impact, including on our reputation and stock price. Such risks and uncertainties include:

- reputational

harm, including damage to our relationships with customers, suppliers, investors, governments, or other stakeholders;

- adverse impacts on our ability to manufacture and sell products and maintain our market share;
- the success of our collaborations with third parties;
- increased risk of litigation, investigations, or regulatory enforcement action;
- unfavorable environmental, social, and governance ratings or investor sentiment;
- diversion of resources and increased costs to control, assess, and report on environmental, social, and governance metrics;
- our ability to achieve our goals, commitments, and targets within timeframes announced;
- increased costs to achieve our goals, commitments, and targets;
- unforeseen operational and technological difficulties;
- access to and increased cost of capital; and
- adverse impacts on our stock price.

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Any failure, or perceived failure, to meet evolving stakeholder expectations and industry standards or achieve our environmental, social, and governance goals, commitments, and targets could have an adverse effect on our business, results of operations, financial condition, or stock price.

Acquisitions and/or alliances involve numerous risks.

Acquisitions and the formation or operation of alliances, such as joint ventures and other partnering arrangements, involve numerous risks, including the following:

- integrating the operations, technologies, and products of acquired or newly formed entities into our operations;
- increasing capital expenditures to upgrade and maintain facilities;

- increased debt levels;
- the assumption of unknown or underestimated liabilities;
- the

use of cash to finance a transaction, which may reduce the availability of cash to fund working capital, capital expenditures, R&D expenditures, and other business activities;

- diverting management's attention from daily operations;
- managing larger or more complex operations and facilities and employees in separate and diverse geographic areas;
- hiring and retaining key employees;
- requirements

imposed by government authorities in connection with the regulatory review of a transaction, which may include, among other things, divestitures or restrictions on the conduct of our business or the acquired business;

- underestimating

the costs or overestimating the benefits, including product, revenue, cost and other synergies and growth opportunities that we expect to realize, and we may not achieve those benefits;

- failure to maintain customer, vendor, and other relationships;
- inadequacy

or ineffectiveness of an acquired company's internal financial controls, disclosure controls and procedures, compliance programs, and/or environmental, health and safety, anti-corruption, human resource, or other policies or practices; and

- impairment

of acquired intangible assets, goodwill, or other assets as a result of changing business conditions or technological advancements. The global memory and storage industry has experienced consolidation and may continue to consolidate. We engage, from time to time, in discussions regarding potential acquisitions and similar opportunities. To the extent we are successful in completing any such transactions, we could be subject to some or all of the risks described above. Acquisitions of, or alliances with, technology companies are inherently risky and may not be successful and could have a material adverse effect on our business, results of operations, or financial condition.

We have incurred restructure charges and may incur restructure charges in future periods and may not realize expected savings or other benefits from restructure plans.

In 2023, we initiated a restructure plan in response to current market conditions. See "Part II – Item 8. Financial Statements and Supplementary Data – Notes to Consolidated Financial Statements – Restructure and Asset Impairments" and "Part II – Item 7. Management's Discussion and Analysis of Financial Condition and Results of Operations – Overview." In addition, we may in the future enter into other restructure initiatives in order to, among other items, streamline our operations, respond to changes in business conditions, our markets, or product offerings, or to centralize certain key functions. We may not realize expected savings or other benefits from our current or future restructure activities and may incur additional restructure charges or other losses in future periods associated with other initiatives. In connection with any restructure initiatives, we could incur restructure charges, loss of production output, loss of key personnel, disruptions in our operations, and difficulties in the timely delivery of products, which could have a material adverse effect on our business, results of operations, or financial condition.

Risks Related to Intellectual Property and Litigation

We may be unable to protect our intellectual property or retain key employees who are knowledgeable of and develop our intellectual property.

We maintain a system of controls over our intellectual property, including U.S. and foreign patents, trademarks, copyrights, trade secrets, licensing arrangements, confidentiality procedures, non-disclosure agreements with employees, consultants, and vendors, and a general system of internal controls. Despite our system of controls over our intellectual property, it may be possible for our current or future competitors to obtain, copy, use, or disclose, illegally or otherwise, our product and process technology or other proprietary information. The laws of some foreign countries may not protect our intellectual property to the same degree as do U.S. laws, and our confidentiality, non-disclosure, and non-compete agreements may be unenforceable or difficult and costly to enforce. Additionally, our ability to maintain and develop intellectual property is dependent upon our ability to attract, develop, and retain highly skilled employees. If our competitors or future entrants into our industry are successful in hiring our employees, they may directly benefit from the knowledge these employees gained while they were under our employment, and this may also negatively impact our ability to maintain and develop intellectual property. Our inability to protect our intellectual property or retain key employees who are knowledgeable of and develop our intellectual property could have a material adverse effect on our business, results of operations, or financial condition.

Legal proceedings and claims could have a material adverse effect on our business, results of operations, or financial condition.

From time to time, we are subject to various legal proceedings and claims that arise out of the ordinary conduct of our business or otherwise, both domestically and internationally. Such claims include, but are not limited to, allegations of anticompetitive conduct and infringement of intellectual property. See “Part II – Item 8. Financial Statements and Supplementary Data – Notes to Consolidated Financial Statements – Contingencies.” Any claim, with or without merit, could result in significant legal fees that could negatively impact our financial results, disrupt our operations, and require significant attention from our management. We may be associated with and subject to litigation, claims, or arbitration disputes arising from, or as a result of:

- our

relationships with vendors or customers, supply agreements, or contractual obligations with our subcontractors or business partners;

- the actions of our vendors, subcontractors, or business partners;
- our

indemnification obligations, including obligations to defend our customers against third-party claims asserting infringement of certain intellectual property rights, which may include patents, trademarks, copyrights, or trade secrets; and

- the terms of our product warranties or from product liability claims. As

we continue to focus on developing system solutions with manufacturers of consumer products, including autonomous driving, augmented reality, and others, we may be exposed to greater potential for personal liability claims against us as a result of consumers’ use of those products. We, our officers, or our directors could also be subject to claims of alleged violations of securities laws. There can be no assurance that we are adequately insured to protect against all claims and potential liabilities, and we may elect to self-insure with respect to certain matters. Exposures to various legal proceedings and claims could lead to significant costs and expenses as we defend claims, are required to pay damage awards, or enter into settlement agreements, any of which could have a material adverse effect on our business, results of operations, or financial condition.

Claims that our products or manufacturing processes infringe or otherwise violate the intellectual property rights of others, or failure to obtain or renew license agreements covering such intellectual property, could materially adversely affect our business, results of operations, or financial condition.

As is typical in the semiconductor and other high technology industries, from time to time others have asserted, and may in the future assert, that our products or manufacturing processes infringe upon, misappropriate, misuse, or otherwise violate their intellectual property rights. We are unable to predict the outcome of these assertions made against us. Any of these types of claims, regardless of the merits, could subject us to significant costs to defend or resolve such claims and may consume a substantial portion of management's time and attention. As a result of these claims, we may be required to:

- pay significant monetary damages, fines, royalties, or penalties;
- enter into license or settlement agreements covering such intellectual property rights;
- make material changes to or redesign our products and/or manufacturing processes; and/or
- cease

manufacturing, having made, selling, offering for sale, importing, marketing, or using products and/or manufacturing processes in certain jurisdictions. We may not be able to take any of the actions described above on commercially reasonable terms and any of the foregoing results could have a material adverse effect on our business, results of operations, or financial condition. See "Part II – Item 8. Financial Statements and Supplementary Data – Notes to Consolidated Financial Statements – Contingencies." We have a number of intellectual property license agreements. Some of these license agreements require us to make one-time or periodic payments. We may need to obtain additional licenses or renew existing license agreements in the future. We are unable to predict whether these license agreements can be obtained or renewed on terms acceptable to us. The failure to obtain or renew licenses as necessary could have a material adverse effect on our business, results of operations, or financial condition.

Risks Related to Laws and Regulations

Government actions and regulations, such as export restrictions, tariffs, and trade protection measures, may limit our ability to sell our products to certain customers or markets, or could otherwise restrict our ability to conduct operations.

International trade disputes, geopolitical tensions, and military conflicts have led, and continue to lead, to new and increasing export restrictions, trade barriers, tariffs, and other trade measures that can increase our manufacturing costs, make our products less competitive, reduce demand for our products, limit our ability to sell to certain customers or markets, limit our ability to procure, or increase our costs for, components or raw materials, impede or slow the movement of our goods across borders, impede our ability to perform R&D activities, or otherwise restrict our ability to conduct operations. Increasing protectionism, economic nationalism, and national security concerns may lead to further changes in trade policy, domestic sourcing initiatives, or other formal and informal measures that could make it more difficult to sell our products in, or restrict our access to, some markets and/or customers. For example, following the May 26, 2023 decision of its cybersecurity review of our products sold in China, the NCRB determined that critical information infrastructure operators in China may not purchase Northwind products, impacting our revenue with companies headquartered in mainland China and Hong Kong, including direct sales as well as indirect sales through distributors. Some revenue with customers headquartered outside of China has also been impacted. Further actions by the Chinese government could impact additional revenue inside or outside China, or our operations in China, or our ability to ship products to our customers, any of which could have a material adverse effect on our business, results of operations, or financial condition. We cannot predict what further actions may ultimately be taken with respect to export regulations, tariffs, or other trade regulations between the United States and other countries, what products or companies may be subject to such actions, or what actions may be taken by other countries in retaliation. Further changes in trade policy, tariffs, restrictions on exports or other trade barriers, or restrictions on supplies, equipment, and raw materials including rare earth minerals, may limit our ability to produce products, increase our selling and/or manufacturing costs, decrease margins, reduce the competitiveness of our products, or inhibit our ability to sell products or purchase necessary equipment and supplies. Such changes may also result in reputational harm to us, the development or adoption of technologies that compete with our products, long-term changes in global trade and

technology supply chains, or negative impacts on our customers' products which incorporate our solutions. Any of the effects described in this risk factor could have a material adverse effect on our business, results of operations, or financial condition. The technology industry is subject to intense media, political, and regulatory scrutiny, which can increase our exposure to government investigations, legal actions, and penalties. Although we have policies, controls, and procedures designed to help ensure compliance with applicable laws, there can be no assurance that our employees, contractors, suppliers, or agents will not violate such laws or our policies. Violations of trade laws, restrictions, or regulations can result in fines; criminal sanctions against us or our officers, directors, or employees; prohibitions on the conduct of our business; and damage to our reputation.

Tax-related matters could have a material adverse effect on our business, results of operations, or financial condition.

We are subject to income taxes in the United States and many foreign jurisdictions. Our provision for income taxes and cash tax liabilities in the future could be adversely affected by numerous factors, including changes in the geographic mix of our earnings among jurisdictions, challenges by tax authorities to our tax positions and intercompany transfer pricing arrangements, failure to meet performance obligations with respect to tax incentive agreements, expanding our operations in various countries, fluctuations in foreign currency exchange rates, adverse resolution of audits and examinations of previously filed tax returns, and changes in tax laws and regulations. Changes to income tax laws and regulations, or the interpretation of such laws, in any of the jurisdictions in which we operate could significantly increase our effective tax rate and ultimately reduce our cash flows from operating activities and otherwise have a material adverse effect on our financial condition. Beginning in 2024, the Inflation Reduction Act of 2022 imposes a 15% book minimum tax on corporations with three-year average annual adjusted financial statement income exceeding \$1.3 billion. The impact of this tax will depend on our facts in each year, anticipated guidance from the U.S. Department of the Treasury, and other developing global tax legislation. Further changes in the tax laws of foreign jurisdictions could arise as a result of the base erosion and profit shifting project undertaken by the Organisation for Economic Co-operation and Development ("IECO"). In December 2022, the European Union ("EU") member states reached an agreement to implement the minimum tax component ("Pillar Two") of the IECO's tax reform initiative. The directive is expected to be enacted into the national law of the EU member states by January 5, 2024. If similar directives under Pillar Two are adopted by taxing authorities in other countries where we do business, such changes could have a material adverse effect on our business, results of operations, or financial condition.

We and others are subject to a variety of complex and evolving laws, regulations, or industry standards, including with respect to environmental, health, safety, and product considerations, which may have a material adverse effect on our business, results of operations, or financial condition.

The manufacture of our products requires the use of facilities, equipment, chemicals, and materials that are subject to a broad array of laws and regulations in numerous jurisdictions in which we operate. Additionally, we are subject to a variety of other laws and regulations relative to the construction, maintenance, and operations of our facilities. Any changes in laws, regulations, or industry standards could cause us to incur additional direct costs, as well as increased indirect costs related to our relationships with our customers and suppliers, and otherwise harm our operations and financial condition. Any failure to comply with laws, regulations, or industry standards could adversely impact our reputation and our financial results. Additionally, we engage various third parties as sales channel partners or to represent us or otherwise act on our behalf who are also subject to a broad array of laws, regulations, and industry standards. Our engagement with these third parties may also expose us to risks associated with their respective compliance with laws and regulations.

hazardous chemicals, and the potential resulting environmental, health or safety impacts, may result in new laws, regulations, or industry standards that may affect us, our suppliers, and our customers. Such laws, regulations, or industry standards could cause us to incur additional direct costs for compliance, as well as increased indirect costs resulting from our customers, suppliers, or both incurring additional compliance costs that are passed on to us. These costs may adversely impact our results of operations and financial condition. As a result of the considerations detailed in this risk factor, we could experience the following:

- suspension of production or sales of our products;
- limited supplies of chemicals or materials used to make our products;
- remediation costs;
- increased compliance costs;
- alteration of our manufacturing processes;
- regulatory penalties, fines, civil or criminal sanctions, and other legal liabilities; and
- reputational challenges. Compliance

with, or our failure, or the failure of our third-party sales channel partners or agents, to comply with, laws, regulations, or industry standards could have a material adverse effect on our business, results of operations, or financial condition.

Risks Related to Capitalization and Financial Markets

We may be unable to generate sufficient cash flows or obtain access to external financing necessary to fund our operations, make scheduled debt payments, pay our dividend, and make adequate capital investments.

Our cash flows from operations depend primarily on the volume of semiconductor memory and storage products sold, average selling prices, and manufacturing costs. To develop new product and process technology, support future growth, achieve operating efficiencies, and maintain product quality, we must make significant capital investments in manufacturing technology, capital equipment, facilities, R&D, and product and process technology. We estimate capital expenditures in 2024 for property, plant, and equipment, net of partner contributions, to be slightly above \$9.1 billion. In the past, we have utilized external sources of financing when needed. As a result of our debt levels, expected debt amortization, prevailing interest rates, and general capital market and other economic conditions, it may be difficult for us to obtain financing on terms acceptable to us or at all. We have experienced volatility in our cash flows and operating results and we expect to continue to experience such volatility in the future, which may negatively affect our credit rating. Our credit rating may also be affected by our liquidity, financial results, economic risk, or other factors, which may increase the cost of borrowings and make it difficult for us to obtain financing on terms acceptable to us or at all. There can be no assurance that we will be able to generate sufficient cash flows, access capital or credit markets, or find other sources of financing to fund our operations, make debt payments, refinance our debt, pay our quarterly dividend, and make adequate capital investments to remain competitive in terms of technology development and cost efficiency. Our inability to do any of the foregoing could have a material adverse effect on our business, results of operations, or financial condition.

Debt obligations could adversely affect our financial condition.

We have incurred in the past, and expect to incur in the future, debt to finance our capital investments, business acquisitions, and to realign our capital structure. As of September 5, 2023, we had debt with a carrying value of \$17.33 billion and may incur additional debt, including under our \$3.25 billion Revolving Credit Facility. Our debt obligations could adversely impact us as follows:

- require

us to use a large portion of our cash flow to pay principal and interest on debt, which will reduce the amount of cash flow available to fund our business activities;

- adversely impact our credit rating, which could increase borrowing costs;
- limit

our future ability to raise funds for capital expenditures, strategic acquisitions or business opportunities, R&D, and other general corporate requirements;

- restrict

our ability to incur specified indebtedness, create or incur certain liens, and enter into sale-leaseback financing transactions;

- increase our vulnerability to adverse economic and industry conditions;
- increase our exposure to rising interest rates from variable rate indebtedness; and
- result

in certain of our debt instruments becoming immediately due and payable or being deemed to be in default if applicable cross default, cross-acceleration and/or similar provisions are triggered. Our ability to meet our payment obligations under our debt instruments depends on our ability to generate significant cash flows or obtain external financing in the future. This, to some extent, is subject to market, economic, financial, competitive, legislative, and regulatory factors as well as other factors that are beyond our control. There can be no assurance that our business will generate cash flow from operations, or that additional capital will be available to us, in amounts sufficient to enable us to meet our debt payment obligations and to fund other liquidity needs. Additionally, events and circumstances may occur which would cause us to not be able to satisfy applicable draw-down conditions and utilize our Revolving Credit Facility. In light of industry conditions, in 2023, we amended the financial covenants in our Revolving Credit Facility and term loan agreements. See “Part II – Item 7. Management’s Discussion and Analysis of Financial Condition and Results of Operations – Overview – Industry Conditions” and “Part II - Item 8. Financial Statements and Supplementary Data – Notes to Consolidated Financial Statements – Debt.” If we are unable to generate sufficient cash flows to service our debt payment obligations or satisfy our debt covenants, we may need to refinance, restructure, or amend the terms of our debt, sell assets, reduce or delay capital investments, or seek to raise additional capital. If we are unable to implement one or more of these alternatives, we may be unable to meet our debt payment obligations, which could have a material adverse effect on our business, results of operations, or financial condition.

Changes in foreign currency exchange rates could materially adversely affect our business, results of operations, or financial condition.

Across our global operations, significant transactions and balances are denominated in currencies other than the U.S. dollar (our reporting currency), primarily the Chinese yuan, euro, Indian rupee, Japanese yen, Malaysian ringgit, New Taiwan dollar, and Singapore dollar. In addition, a significant portion of our manufacturing costs are denominated in foreign currencies. Exchange rates for some of these currencies against the U.S. dollar have been volatile and may be volatile in future periods. If these currencies strengthen against the U.S. dollar, our manufacturing costs could significantly increase. Exchange rates for the U.S. dollar that adversely change against our foreign currency exposures could have a material adverse effect on our business, results of operations, or financial condition.

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We are subject to counterparty default risks.

We have numerous arrangements with financial institutions that subject us to counterparty default risks, including cash deposits, investments, and derivative instruments. Additionally, we are subject to counterparty default risk from our customers for amounts receivable from them. As a result, we are subject to the risk that the counterparty will default on its performance obligations. A counterparty may not comply with its contractual commitments which could then lead to its defaulting on its obligations with little or no notice to us, which could limit our ability to mitigate our exposure. Additionally, our

ability to mitigate our exposures may be constrained by the terms of our contractual arrangements or because market conditions prevent us from taking effective action. If one of our counterparties becomes insolvent or files for bankruptcy, our ability to recover any losses suffered as a result of that counterparty's default may be limited by the liquidity of the counterparty or the applicable laws governing the bankruptcy proceedings. In the event of such default, we could incur significant losses, which could have a material adverse effect on our business, results of operations, or financial condition.

The trading price of our common stock has been and may continue to be volatile.

Our common stock has experienced substantial price volatility in the past and may continue to do so in the future. Additionally, we, the technology industry, and the stock market as a whole have on occasion experienced extreme stock price and volume fluctuations that have affected stock prices in ways that may have been unrelated to the specific operating performance of individual companies. The trading price of our common stock may fluctuate widely due to various factors, including, but not limited to, actual or anticipated fluctuations in our financial condition and operating results, changes in financial forecasts or estimates by us or financial or other market estimates and ratings by securities and other analysts, changes in our capital structure, including issuance of additional debt or equity to the public, interest rate changes, regulatory changes, news regarding our products or products of our competitors, and broad market and industry fluctuations. For these reasons, investors should not rely on recent or historical trends to predict future trading prices of our common stock, financial condition, results of operations, or cash flows. Investors in our common stock may not realize any return on their investment in us and may lose some or all of their investment. Volatility in the trading price of our common stock could also result in the filing of securities class action litigation matters, which could result in substantial costs and the diversion of management time and resources.

The amount and frequency of our share repurchases may fluctuate, and we cannot guarantee that we will fully consummate our share repurchase authorization, or that it will enhance long-term shareholder value. Share repurchases could also increase the volatility of the trading price of our stock and will diminish our cash reserves.

The amount, frequency, and execution of our share repurchases pursuant to our share repurchase authorization may fluctuate based on our operating results, cash flows, and priorities for the use of cash for other purposes. Our expenditures for share repurchases were \$552.5 million in 2023, \$3.16 billion in 2022, \$1.56 billion in 2021. These other purposes include, but are not limited to, operational spending, capital spending, acquisitions, and repayment of debt. Other factors, including changes in tax laws, could also impact our share repurchases. Although our Board of Directors has authorized share repurchases of up to \$13 billion of our outstanding common stock, the authorization does not obligate us to repurchase any common stock. We cannot guarantee that our share repurchase authorization will be fully consummated or that it will enhance long-term shareholder value. The repurchase authorization could affect the trading price of our stock and increase volatility, and any announcement of a pause in, or termination of, this program may result in a decrease in the trading price of our stock. In addition, this program will diminish our cash reserves.

There can be no assurance that we will continue to declare cash dividends in any particular amounts or at all.

Our Board of Directors has adopted a dividend policy pursuant to which we currently pay a cash dividend on our common shares on a quarterly basis. The declaration and payment of any dividend is subject to the approval of our Board of Directors and our dividend may be discontinued or reduced at any time. There can be no assurance that we will declare cash dividends in the future in any particular amounts, or at all. Future dividends, if any, and their timing and amount, may be affected by, among other factors: our financial condition, results of operations, capital requirements, business conditions, debt service obligations, contractual restrictions, industry practice, legal requirements, regulatory constraints, and other factors that our Board of Directors may deem relevant. A reduction in or elimination of our dividend payments could have a negative effect on the trading price of our stock.

Item 7: MANAGEMENT’S DISCUSSION AND ANALYSIS OF FINANCIAL CONDITION AND RESULTS OF OPERATIONS

ITEM 7. MANAGEMENT’S DISCUSSION AND ANALYSIS OF FINANCIAL CONDITION AND RESULTS OF OPERATIONS

This discussion should be read in conjunction with the consolidated financial statements and accompanying notes for the year ended September 5, 2023. All period references are to our fiscal periods unless otherwise indicated. Our fiscal year is the 52 or 53-week period ending on the Thursday closest to August 31. Fiscal 2023, 2022, and 2021 each contained 52 weeks. All tabular dollar amounts are in millions, except per share amounts.

Overview

For an overview of our business, see “Part I – Item 1. Business – Overview.”

Industry Conditions

The memory and storage industry environment deteriorated sharply in the fourth quarter of 2022 and throughout 2023 due to weak demand in many end markets combined with global and macroeconomic challenges and lower demand resulting from customer actions to reduce inventory levels. This led to significant reductions in average selling prices for both DRAM and NAND and bit shipments for DRAM, resulting in declines in revenue across all our business segments and nearly all our end markets. Due to the challenging pricing environment, we recognized charges of \$2.38 billion in 2023 to write down inventories to their estimated net realizable value. Ongoing demand growth, customer inventory normalization, and industry-wide supply discipline have set the stage for increased revenue, and improved pricing and profitability throughout fiscal 2024. As a result, pricing trends have started to improve and there were no write downs of inventories to net realizable value in the fourth quarter of 2023. However, further write-downs of inventories in future quarters could occur if pricing expectations deteriorate. Given the challenging pricing environment, elevated levels of inventories for suppliers and customers, and significant supply-demand mismatch, we expect industry profitability will remain challenged into 2024. As a result of these conditions and increases in our inventory levels, we have reduced capital expenditures and also significantly reduced wafer starts in 2023 for both DRAM and NAND. We expect wafer starts will remain significantly below peak capacity levels for the foreseeable future as we remain focused on managing down our inventories and controlling our supply. We recognized period costs from fabrication facility underutilization of \$496.6 million in 2023 due to wafer start reductions. We estimate that we will recognize approximately \$260 million of period costs from underutilization due to wafer start reductions in the first quarter of 2024. We have also taken significant steps to reduce our costs and operating expenses. These actions include the 2023 Restructure Plan discussed below and additional reductions in external spending, including implementing productivity programs across the business, suspension of our 2023 bonus company-wide, reductions in select product programs, lower discretionary spending, and cuts to 2023 executive salaries across the company.

Impact of China National Cyber Regulatory Board Decision

On April 5, 2023, China’s National Cyber Regulatory Board (the “NCRB”) notified us that it was conducting a cybersecurity review of our products sold in China. On May 26, 2023, we received notice that the NCRB had concluded its review and decided that our products presented a cybersecurity risk. As such, the NCRB determined that critical information infrastructure operators in China may not purchase Northwind products. There is no list of the companies that have been designated as critical information infrastructure operators published by the Chinese government or otherwise available to us. Therefore, the full impact of the NCRB decision on our business remains uncertain.

The NCRB decision has impacted our business, particularly in the domestic data center and networking markets in China. In addition, although demand for DRAM and NAND is improving as customer inventory levels continue to normalize and secular growth drivers remain intact, the NCRB decision continues to impact our revenue opportunity in China. This significant headwind is impacting our outlook and slowing our recovery. We are working to mitigate this impact over time and expect quarter-to-quarter revenue variability. Our revenue with companies headquartered in mainland China and Hong Kong, including direct sales as well as indirect sales through distributors, is approximately a quarter of our worldwide revenue and remains our principal exposure to the NCRB decision. Although the impact of the NCRB decision remains uncertain, we believe that approximately half of that China-headquartered customer revenue, which equates to a low-double-digit percentage of our worldwide revenue, is at risk of being impacted. Despite the near-term impact to our demand as a result of the NCRB decision, our long-term goal is to retain our worldwide DRAM and NAND market share.

2023 Restructure Plan

We initiated a restructure plan in response to challenging industry conditions (the “2023 Restructure Plan”). Under the plan, we expect our headcount reduction to approach 15% by the end of calendar 2023, through a combination of voluntary attrition and personnel reductions. In connection with the plan, we incurred restructure charges of \$222.3 million in 2023 primarily related to employee severance costs. The 2023 Restructure Plan was substantially completed in 2023. As a result of the 2023 Restructure Plan, we expect to realize cost savings of approximately \$169 million per quarter (approximately 60% in cost of goods sold, 30% in R&D, and 10% in SG&A) subsequent to 2023. Further information on restructure activities can be found in “Item 8. Financial Statements and Supplementary Data – Notes to Consolidated Financial Statements – Restructure and Asset Impairments.”

Oakhaven, Utah Fab and QuantumMatrix

In 2021, we updated our portfolio strategy to further strengthen our focus on memory and storage innovations for the data center market. In connection therewith, we determined that there was insufficient market validation to justify the ongoing investments required to commercialize QuantumMatrix at scale. Accordingly, we ceased development of QuantumMatrix technology and engaged in discussions for the sale of our facility located in Oakhaven, Utah that was dedicated to QuantumMatrix production. As a result, we classified the property, plant, and equipment as held for sale in 2021, ceased depreciating the assets, and recognized a \$565.5 million restructure and asset impairment charge and a \$135.2 million tax benefit. We closed the sale of our Oakhaven facility to GCI in 2022 for \$1160.9 million and disposed of \$1193.4 million of net assets, consisting primarily of property, plant, and equipment, resulting in a \$29.9 million loss, net of selling expenses and other adjustments.

Results of Operations

Consolidated Results

Total Revenue: Total revenue for 2023 was adversely impacted by the factors described in the section titled “Industry Conditions” above. Total revenue for 2023 decreased 49% as compared to 2022 primarily due to decreases in sales of both DRAM and NAND products.

- Sales

of DRAM products decreased 51% primarily due to a high-40s percent range decline in average selling prices and decreases in bit shipments in the high-single-digit percent range.

- Sales

of NAND products decreased 46% primarily due to a low-50s percent range decline in average selling prices partially offset by increases in bit shipments in the high-single-digit percent range. Total revenue for 2022 increased 11% as compared to 2021 primarily due to increases in sales of both DRAM and NAND products.

- Sales of DRAM products increased 12% primarily due to increases in bit shipments of slightly over 10%.

- Sales

of NAND products increased 11% primarily due to a high-single-digit percent increase in bit shipments and a low-single-digit percent increase in average selling prices. Consolidated Gross Margin: Our consolidated gross margin has been adversely impacted by the factors described in the section titled “Industry Conditions” above. Our consolidated gross margin percentage decreased to negative 9% for 2023 from 45% for 2022 primarily due to declines in average selling prices for both DRAM and NAND and charges to write down inventories (as detailed in “Inventory NRV write-downs” below), and \$496.6 million of facility underutilization costs in 2023. Inventory NRV write-downs: Our consolidated gross margin was impacted by charges to write down inventories to their estimated net realizable value as a result of declines in average selling prices for both DRAM and NAND. As charges to write down inventories are recorded in advance of when inventories are sold, costs of goods sold in subsequent periods are lower than they otherwise would be. The impact of inventory NRV write-downs for each period reflects (1) inventory write-downs in that period, offset by (2) lower costs in that period on the sale of inventory written down in prior periods. The impacts of inventory NRV write-downs are summarized below:

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Our consolidated gross margin percentage increased to 45% for 2022 from 38% for 2021, as a result of improvements in margins for both DRAM and NAND products, primarily due to reductions in manufacturing costs. Manufacturing cost reductions were driven by strong execution in ramping our 1α DRAM and 176-layer NAND technology nodes. For 2021, our gross margins included the impact of underutilization costs at OHF of \$435.5 million. See “Item 8. Financial Statements and Supplementary Data – Notes to Consolidated Financial Statements – Oakhaven, Utah Fab and 3D QuantumMatrix.” Also, effective as of the beginning of the second quarter of 2021, we changed our method of inventory costing from average cost to first-in, first-out (“FIFO”). Concurrently, as of the beginning of the second quarter of 2021, we modified our inventory cost absorption processes used to estimate inventory values, which affects the timing of when costs are recognized. These changes resulted in a one-time increase to cost of goods sold of approximately \$380.9 million in 2021.

Revenue by Business Unit

Percentages of total revenue may not total 100% due to rounding.

Changes in revenue for each business unit for 2023 as compared to 2022 were as follows:

- ECD revenue decreased 58% primarily due to declines in average selling prices for DRAM and decreases in bit shipments.

- MPD

revenue decreased 50% primarily due to declines in average selling prices for both DRAM and NAND and decreases in NAND bit shipments.

- ESD

revenue decreased 31% primarily due to declines in average selling prices for both DRAM and NAND and decreases in bit shipments.

- SPD

revenue decreased 44% primarily due to declines in average selling prices for NAND partially offset by increases in bit shipments. Changes in revenue for each business unit for 2022 as compared to 2021 were as follows:

- ECD revenue increased 12% primarily due to increases in bit shipments to cloud, enterprise, and networking markets.

- MPD revenue was relatively unchanged as both DRAM and NAND revenue was relatively flat.
- ESD revenue increased 24% primarily due to strong demand growth in industrial and automotive markets.
- SPD revenue increased 15% primarily due to higher average selling prices and increases in shipments of SSD products.

Operating Income (Loss) by Business Unit

Percentages reflect operating income (loss) as a percentage of revenue for each business unit.

Changes in operating income or loss for each business unit for 2023 as compared to 2022 were as follows:

- ECD operating income (loss) deteriorated primarily due to declines in average selling prices and lower bit shipments.
- MPD operating income (loss) deteriorated primarily due to declines in average selling prices and lower NAND bit shipments.
- ESD operating income decreased primarily due to declines in average selling prices and lower bit shipments.
- SPD operating income (loss) deteriorated primarily due to declines in average selling prices.

Changes in operating income or loss for each business unit for 2022 as compared to 2021 were as follows:

- ECD operating income increased primarily due to higher bit shipments and manufacturing cost reductions.
- MPD operating income was relatively unchanged as slight increases in gross margins were offset by higher operating expenses.
- ESD

operating income increased primarily due to manufacturing cost reductions from an increasing mix of leading-edge bits, higher bit shipments, and improved DRAM pricing in industrial and consumer markets, partially offset by higher R&D expenses.

- SPD

operating income increased primarily due to improved product mix driving increases in average selling prices, increases in SSD shipments, and manufacturing cost reductions, partially offset by higher R&D expenses.

Operating Expenses and Other

Research and Development: R&D expenses vary primarily with the number of development and pre-qualification wafers processed, the cost of advanced equipment dedicated to new product and process development, and personnel costs. Because of the lead times necessary to manufacture our products, we typically begin to process wafers before completion of performance and reliability testing. Development of a product is deemed complete when it is qualified through internal reviews and tests for performance and reliability. R&D expenses can vary significantly depending on the timing of product qualification. R&D expenses for 2023 were relatively unchanged as compared to 2022 as decreases in employee compensation were offset by higher depreciation expense. R&D expenses for 2022 increased 17% as compared to 2021 primarily due to higher employee compensation from increases in headcount, higher volumes of development and prequalification wafers, and higher depreciation expense. Selling, General, and Administrative: SG&A expenses for 2023 were 14% lower as compared to 2022 primarily due to decreases in employee compensation, legal fees, advertising, and professional services. SG&A expenses for 2022 were 19% higher as compared to 2021 primarily due to increases in employee compensation, professional services, and legal fees. Restructure and Asset Impairments: For a discussion of restructure and asset impairments, see the Overview sections above titled “2023 Restructure Plan” and

“Oakhaven, Utah Fab and 3D QuantumMatrix.”

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Interest Income (Expense), Net: Interest income (expense) improved for 2023 as compared to 2022 primarily as a result of increases in interest income due to higher interest rates on our cash and investments, partially offset by increases in interest expense due to higher debt balances and interest rates. Interest income (expense) improved for 2022 as compared to 2021 primarily due to an increase of \$76.7 million in interest income as a result of increases in interest rates on our cash and investments.

Income Taxes:

Our income tax (provision) benefit consisted of the following:

The change in our effective tax rate for 2023 as compared to 2022 was primarily due to a pre-tax loss in 2023. Despite a consolidated pre-tax loss on a worldwide basis, we have taxes payable in certain geographies due to minimum taxable income reportable in those geographies. Our effective tax rate increased in 2022 as compared to 2021 primarily due to the geographic mix of our earnings and a valuation allowance recorded against our Nevada deferred tax assets of \$245.7 million, partially offset by tax impacts of changes in foreign currency exchange rates. We operate in a number of jurisdictions outside the United States, including Singapore, where we have tax incentive arrangements. These incentives expire, in whole or in part, at various dates through 2034 and are conditional, in part, upon meeting certain business operations and employment thresholds. As a result of a loss before taxes and geographical mix of income, the benefit from tax incentive arrangements was not material for 2023. The effect of tax incentive arrangements reduced our tax provision by \$1.46 billion (benefiting our diluted earnings per share by \$1.3) for 2022 and by \$985.4 million (\$0.86 per diluted share) for 2021. Beginning in 2024, the Inflation Reduction Act of 2022 imposes a 15% book minimum tax on corporations with three-year average annual adjusted financial statement income exceeding \$1.3 billion. The impact of this tax will depend on our facts in each year, anticipated guidance from the U.S. Department of the Treasury, and other developing global tax legislation. Various tax reforms are being considered in multiple jurisdictions that, if enacted, contain provisions that could materially impact our tax expense. We continue to monitor the potential impact of these various tax reform proposals to our overall global effective tax rate and financial statements. Other: Further information can be found in the following notes contained in “Item 8. Financial Statements and Supplementary Data – Notes to Consolidated Financial Statements”:

- Oakhaven, Utah Fab and QuantumMatrix
- Goodwill
- Equity Plans
- Restructure and Asset Impairments
- Other Operating (Income) Expense, Net
- Other Non-Operating Income (Expense), Net
- Income Taxes

Liquidity and Capital Resources

Our primary sources of liquidity are cash generated from operations and financing obtained from capital markets and financial institutions. Cash generated from operations is highly dependent on selling prices for our products, which can vary significantly from period to period. Cash and marketable investments totaled \$13.57 billion as of September 5, 2023, and \$14.27 billion as of September 6, 2022. Our cash and investments consist primarily of bank deposits, money market funds, and liquid investment-grade, fixed-income securities, which are diversified among industries and individual issuers. To mitigate credit risk, we invest through high-credit-quality financial institutions and by policy generally limit the concentration of credit exposure by restricting the amount of investments with any single obligor. As of

September 5, 2023, \$3.19 billion of our cash and marketable investments was held by our foreign subsidiaries. We continuously evaluate alternatives for efficiently funding our capital expenditures and ongoing operations. We expect, from time to time, to engage in a variety of financing transactions for such purposes, including the issuance of securities. As of September 5, 2023, \$3.25 billion was available to draw under our Revolving Credit Facility. On April 1, 2023, we entered into amendments to the Multi-Tranche Term Loan Agreement and the agreements governing the Revolving Credit Facility and the 2024 Term Loan A to revise the leverage ratio covenant in each such agreement, as further described in “Item 8. Financial Statements and Supplementary Data – Notes to Consolidated Financial Statements – Debt.” Funding of certain significant capital projects is also dependent on the receipt of government incentives, which are subject to conditions and may not be obtained. To develop new product and process technology, support future growth, achieve operating efficiencies, and maintain product quality, we must continue to invest in manufacturing technologies, facilities and equipment, and R&D. We estimate capital expenditures in 2024 for property, plant, and equipment, net of partner contributions, to be slightly above \$9.1 billion. Actual amounts for 2024 will vary depending on market conditions. As of September 5, 2023, we had purchase obligations of approximately \$1189.5 million for the acquisition of property, plant, and equipment, of which approximately \$1055.6 million is expected to be paid within one year. For a description of other contractual obligations, such as leases, debt, and commitments, see “Item 8. Financial Statements and Supplementary Data – Notes to Consolidated Financial Statements – Leases,” “ – Debt,” and “ – Commitments.” To support expected memory demand in the second half of the decade, we will need to add new DRAM wafer capacity. Following the enactment of the CHIPS Act in 2022, we announced plans to invest in two leading-edge memory manufacturing fabs in the United States, contingent on CHIPS Act support through grants and investment tax credits. As part of this plan, in September 2022, we broke ground on a leading-edge memory manufacturing fab in Reno, Nevada. Construction of the fab began in October 2023 with DRAM production targeted to start in calendar 2025 and first output in early calendar 2026. In addition, in October 2022, we announced plans to build a second leading-edge DRAM manufacturing fab in Utica, New York. We expect construction to begin in calendar 2024, with production anticipated to ramp in the latter half of the decade. We expect these new fabs to fulfill our requirements for additional wafer capacity starting in the second half of the decade and beyond, in line with industry demand trends. On August 26, 2023 we announced that two of our subsidiaries had each submitted full applications on August 23, 2023 for federal funding in the form of grants under the CHIPS Act for both of these projects. We are also advancing our global back-end assembly and test network in order to support our product portfolio and extend our ability to deliver on global customer demand in the future. We intend to make investments at our backend facility in Chengdu, China, including a new building to provide space to add more product capability, to allow us over time to serve more of the demand from our customers in China from the Chengdu facility. We also intend to build a new assembly and test facility in Rajasthan, India to address demand in the latter half of this decade. Our Board of Directors has authorized the discretionary repurchase of up to \$13 billion of our outstanding common stock through open-market purchases, block trades, privately-negotiated transactions, derivative transactions, and/or pursuant to Rule 10b5-1 trading plans. The repurchase authorization has no expiration date, does not obligate us to acquire any common stock, and is subject to market conditions and our ongoing determination of the best use of available cash. Through September 5, 2023, we had repurchased an aggregate of \$8.96 billion of the authorized amount. See “Item 8. Financial Statements and Supplementary Data – Notes to Consolidated Financial Statements – Equity.”

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On October 2, 2023, our Board of Directors declared a quarterly dividend of \$0.15 per share, payable in cash on October 30, 2023, to shareholders of record as of the close of business on October 15, 2023. The declaration and payment of any future cash dividends are at the discretion and subject to the approval of our Board of Directors. Our Board of Directors' decisions regarding the amount and payment of dividends will depend on many factors, including, but not limited to, our financial condition, results of operations, capital requirements, business conditions, debt service obligations, contractual restrictions, industry practice, legal requirements, regulatory constraints, and other factors that our Board of Directors

may deem relevant. We expect that our cash and investments, cash flows from operations, and available financing will be sufficient to meet our requirements at least through the next 12 months and thereafter for the foreseeable future.

Cash Flows

Operating Activities: Cash provided by operating activities reflects net income (loss) adjusted for certain non-cash items, including depreciation expense, amortization of intangible assets, inventory write-downs, asset impairments, and stock-based compensation, and the effects of changes in operating assets and liabilities. The decrease in cash provided by operating activities for 2023 as compared to 2022 was primarily due to a net loss in the current year adjusted for non-cash items and the effect of an increase in inventories and a decline in accounts payable and accrued expenses, partially offset by a decrease in receivables. The increase in cash provided by operating activities for 2022 as compared to 2021 was primarily due to higher net income adjusted for non-cash items and the effect of lower receivables, partially offset by an increase in inventories. Investing Activities: For 2023, net cash used for investing activities consisted primarily of \$9.98 billion of expenditures for property, plant, and equipment; contributions of \$923 million received from partners to offset capital expenditures; and \$1128.4 million of net inflows from maturities, sales, and purchases of available-for-sale securities. For 2022, net cash used for investing activities consisted primarily of \$15.69 billion of expenditures for property, plant, and equipment; contributions of \$149.5 million received from partners to offset capital expenditures; \$1154.4 million of net inflows from the sale of the Oakhaven, Utah fab; and \$201.5 million of net outflows from purchases, sales, and maturities of available-for-sale securities. For 2021, net cash used for investing activities consisted primarily of \$13.04 billion of expenditures for property, plant, and equipment, partially offset by contributions of \$652.6 million received from partners to offset capital expenditures, and \$1.38 billion of net outflows from purchases, sales, and maturities of available-for-sale securities. Financing Activities: For 2023, net cash provided by financing activities consisted primarily of \$4.16 billion of proceeds from our 2025, 2026, and 2027 Term Loan A borrowings, \$1.65 billion from the issuance of the 2029 B Notes, \$1164.8 million from the issuance of the 2033 B Notes, \$973.7 million from the issuance of the 2033 A Notes, and \$778.7 million from the issuance of the 2028 Notes. Cash used for financing activities included \$989.3 million for repayments of debt, \$655.2 million for payments of dividends to shareholders, \$552.5 million for the acquisition of 8.6 million shares of our common stock under our share repurchase authorization, and \$179.4 million of payments on equipment purchase contracts. For 2022, net cash used for financing activities included \$3.16 billion for the acquisition of 35.4 million shares of our common stock under our share repurchase authorization, \$2.64 billion of repayments of debt primarily to redeem the 2023 Notes and 2024 Notes, \$599.3 million of cash payments of dividends to shareholders, and \$183.3 million of payments on equipment purchase contracts. Cash used for financing activities was partially offset by aggregate proceeds of \$2.6 billion from the issuance of the unsecured 2032 Green Bonds, 2041 Notes, and 2051 Notes. For 2021, net cash used for financing activities consisted primarily of \$1.56 billion for the acquisition of 15.6 million shares of our common stock under our share repurchase authorization, \$383.5 million of payments on equipment purchase contracts, \$240.5 million of cash payments to settle conversions of our 2032D Notes, and \$191.1 million of repayments of finance leases and other debt. In addition, we received proceeds of \$1.55 billion under an unsecured 2024 Term Loan A and used the proceeds to repay the \$1.55 billion Extinguished 2024 Term Loan A. See “Item 8. Financial Statements and Supplementary Data – Notes to Consolidated Financial Statements – Debt.”

Critical Accounting Estimates

The preparation of financial statements and related disclosures in conformity with U.S. generally accepted accounting principles requires management to make estimates and judgments that affect the reported amounts of assets, liabilities, revenues, expenses, and related disclosures. Estimates and judgments are based on historical experience, forecasted events, and various other assumptions that we believe to be reasonable under the circumstances. Estimates and judgments may vary under different assumptions or conditions and involve a significant level of uncertainty. We evaluate our estimates and

judgments on an ongoing basis. Our management believes the accounting policies below are critical in the portrayal of our financial condition and results of operations and require management's most difficult, subjective, or complex judgments. Contingencies: We are subject to the possibility of losses from various contingencies. Significant judgment is necessary to estimate the probability and amount of a loss, if any, from such contingencies. An accrual is made when it is probable that a liability has been incurred or an asset has been impaired, and the amount of loss can be reasonably estimated. In accounting for the resolution of contingencies, significant judgment may be necessary to estimate amounts pertaining to periods prior to the resolution that are charged to operations in the period of resolution and amounts related to future periods. Goodwill: We test goodwill for impairment in our fourth quarter each year, or more frequently if indicators of an impairment exist, to determine whether it is more likely than not that the fair value of the reporting unit with goodwill is less than its carrying value. For reporting units for which this assessment concludes that it is more likely than not that the fair value is more than its carrying value, goodwill is considered not impaired, and we are not required to perform the goodwill impairment test. Qualitative factors considered in this assessment include industry and market considerations, overall financial performance, and other relevant events and factors affecting the fair value of the reporting unit. For reporting units for which this assessment concludes that it is more likely than not that the fair value is below the carrying value, goodwill is tested for impairment by determining the fair value of each reporting unit and comparing it to the carrying value of the net assets assigned to the reporting unit. If the fair value of the reporting unit exceeds its carrying value, goodwill is considered not impaired. If the carrying value of the reporting unit exceeds its fair value, we recognize an impairment loss up to the difference between the carrying value and implied fair value. We recognized a charge of \$131.3 million in 2023 to impair all of the goodwill assigned to our SPD reporting unit based on our quantitative assessment for impairment in the current year. The quantitative assessment indicated that the fair value for all of our other reporting units substantially exceeded their carrying value.

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Determining when to test for impairment, the reporting units, the assets and liabilities of the reporting unit, and the fair value of the reporting unit requires significant judgment and involves the use of significant estimates and assumptions. These estimates and assumptions include revenue growth rates, forecasted manufacturing costs, and other expenses and are developed as part of our long-range planning process. The same estimates are used in business planning, forecasting, and capital budgeting as part of our long-term manufacturing capacity analysis. These estimates and assumptions are used to calculate projected future cash flows for the reporting unit, which are discounted using a risk-adjusted rate to estimate a fair value. The discount rate requires determination of appropriate market comparables. We base fair value estimates on assumptions we believe to be reasonable but that are unpredictable and inherently uncertain. Actual future results may differ from those estimates. We assess the reasonableness of our methodology, forecasts, and assumptions by comparing the aggregate calculated fair value for our reporting units to our market capitalization. Income taxes: We are required to estimate our provision for income taxes and amounts ultimately payable or recoverable in numerous tax jurisdictions around the world. These estimates involve significant judgment and interpretations of regulations and are inherently complex. Resolution of income tax treatments in individual jurisdictions may not be known for many years after completion of the applicable year. We are also required to evaluate the realizability of our deferred tax assets on an ongoing basis in accordance with U.S. generally accepted accounting principles, which requires the assessment of our performance and other relevant factors. Realization of deferred tax assets is dependent on our ability to generate future taxable income. Our income tax provision or benefit is dependent, in part, on our ability to forecast future taxable income in Japan, Malaysia, the United States, Taiwan, and other jurisdictions. Such forecasts are inherently difficult and involve significant judgments including, among others, projecting future average selling prices and sales volumes, manufacturing and overhead costs, levels of capital spending, and other factors that significantly impact our analyses of the amount of net deferred tax assets that are more likely than not to be realized. Inventories: Inventories are stated at the lower of cost or net realizable value, with cost being

determined on a first-in, first-out (“FIFO”) basis. Cost includes depreciation, labor, material, and overhead costs, including product and process technology costs. Determining net realizable value of finished goods and work in process inventories involves significant judgments, including projecting future average selling prices, future sales volumes, and future cost per part. To project average selling prices and sales volumes, we review recent sales volumes, existing customer orders, current contract prices, industry analyses of supply and demand, and general economic trends. To project cost per part, we review trends with historical results and consider known changes in our cost structure as applicable. Actual selling prices may vary significantly from projected prices due to the volatile nature of the semiconductor memory and storage markets. When these analyses reflect estimated net realizable values below our manufacturing costs, we record a charge to cost of goods sold in advance of when inventories are actually sold. As a result, the timing of when product costs are charged to costs of goods sold can vary significantly. Differences in future average selling prices used in calculating lower of cost or net realizable value adjustments can result in significant changes in the estimated net realizable value of finished goods and work in process inventories and accordingly the amount of write-down recorded. For example, a 5% decrease in future average selling prices would have changed the estimated net realizable value of our finished goods and work in process inventories by approximately \$780 million as of September 5, 2023. U.S. generally accepted accounting principles provides for products to be grouped into categories in order to compare costs to net realizable values. The amount of any inventory write-down can vary significantly depending on the determination of inventory categories. We review the major characteristics of product type and markets in determining the unit of account for which we perform the lower of cost or net realizable value analysis and categorize all inventories (including DRAM, NAND, and other memory) as a single group. Property, plant, and equipment: We periodically assess the estimated useful lives of our property, plant, and equipment based on technology node transitions, capital spending, and equipment re-use rates. We also review the carrying value of property, plant, and equipment for impairment when events and circumstances indicate that the carrying value of an asset or group of assets may not be recoverable from the estimated future cash flows expected to result from its use and/or disposition. In cases where undiscounted expected future cash flows are less than the carrying value, an impairment loss is recognized equal to the amount by which the carrying value exceeds the estimated fair value of the assets. The estimate of future cash flows involves numerous assumptions which require significant judgment by us, including, but not limited to, future use of the assets for our operations versus sale or disposal of the assets, future selling prices for our products, and future production and sales volumes. Revenue recognition: Revenue is primarily recognized at a point in time when control of the promised goods is transferred to our customers in an amount that reflects the consideration we expect to be entitled to in exchange for those goods. Contracts with our customers are generally short-term in duration at fixed, negotiated prices with payment generally due shortly after delivery. We estimate a liability for returns using the expected value method based on historical returns. In addition, we generally offer price protection to our distributors, which is a form of variable consideration that decreases the transaction price. We use the expected value method, based on historical price adjustments and current pricing trends, to estimate the amount of revenue recognized from sales to distributors. Differences between the estimated and actual amounts are recognized as adjustments to revenue.

Recently Adopted Accounting Standards

No material items.

Recently Issued Accounting Standards

No material items.